Automating Systematic Reviews: A Systematic Review of 128 AI-based Tools for PRISMA-Compliant Systematic Reviews and Meta-Analyses

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Background: Systematic literature reviews (SLRs) and meta-analyses are critical methods for synthesizing research findings, but they are highly labor-intensive and time-consuming. AI technologies, particularly machine learning and deep learning, present an opportunity to automate significant portions of these processes, potentially accelerating the review timeline and reducing human error.

Objective: This study aims to provide a comprehensive overview of AI-based tools and frameworks that support the PRISMA (2020) SLR and meta-analysis processes.

Methods: A total of 128 unique AI-based tools were identified that support different phases of systematic reviews and meta-analyses. The tools were categorized according to the PRISMA (2020) framework and available performance metrics were extracted to assess their effectiveness in automating various steps of the review process.

Results: AI-based tools were found to be most prevalent in the Screening (79.7%) and Identification (53.1%) phases, with fewer tools addressing Data Extraction (37.5%), Data Synthesis (17.2%), Quality Assessment (5.5%), and Bias Assessment (7.0%). Notably 47.7% of tools supported multiple phases with most common overlaps in the combinations of Identification + Screening (40.6%) and Screening + Data Extraction (24.2%). Unsurprisingly Quality Assessment tools showed mostly overlaps with Data Extraction (71.4%) and Bias Assessment (57.1%).

Conclusion: The findings suggest a growing trend in AI tool development focused on the early phases of the systematic review process, especially Identification and Screening. Moreover, a broad spectrum of performance metrics have been used to assess the performance of said AI-based tools. However, more work is needed to evaluate these tools across the board as there is little to no consistency on how these tools are tested (different databases, versions, tools etc.). Lastly more work is needed for the later stages of review, such as Data Synthesis and Quality Assessment. Future research should focus on integrating AI solutions across the entire systematic review process and create a standardized testing methodology.

Additional Key Words and Phrases: AI-based tools, tools, systematic reviews, meta-analysis, automation, artificial intelligence, prisma, machine learning

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1 INTRODUCTION

As Rowley et al. [87] puts it eloquently "Scientific knowledge is based on the accumulated findings from prior research in a research domain, where individual studies constitute building blocks". In many domains it is commonly understood that new studies will build on previous research [33].However, there has been a significant rise in both education researchers and actual academic publications. From 1980 to 2018 the amount of education researchers increased from 4 million to 13.1 million worldwide (3.3-fold increase) [98]. Meanwhile the number of academic publications increased from 0.65 million in 1980 to 3.16 million in 2018 (4.9-fold increase) [98]. This presents a challenge to researchers meaning to build upon the existing knowledge in their research domains. It is simply not feasible for researchers to have to go through the scientific body of their research domain each time.

1.1 Literature Reviews and Meta-Analyses

Systematic literature reviews (SLRs) and Meta-Analyses were developed precisely to address the need for structured evidence aggregation. Systematic literature reviews (SLRs) provide researches with an overview of a research domain in a single paper [76]. Similarly meta-analyses offer the opportunity to critically evaluate and statistically combine results of comparable studies or trials[25]. The latter plays a crucial role in evidence-based research by synthesizing findings from numerous studies and establishing statistical significance with studies that have conflicting results.

1.2 Review Complexitiy and PRISMA 2020

Both SLRs and Meta-analysis are very labor intensive and take long times to complete. Based on the PRISMA framework, which is widely endorsed, they require research questions, search strategy, study selection, data extraction and the synthesis of results [57]. The latest updated guideline of PRISMA 2020 framework comprises of a 27-item checklist [75]. These 27 items are distributed across key phases, including: title and abstract formulation, rationale and objectives, methodology (search strategy, selection process and eligibillity criteria), results (study selection, characteristics and synthesis), discussion and registration and protocol adherence.

In the identification phase, information sources (PRISMA Item 6) and the search strategy (Item 7) describe how and where studies are sourced. The screening phase is guided by the selection process (Item 8), the results of study selection via flow diagrams (Item 16a), and the excluded studies with rationale (Item 16b). Data extraction is addressed through data collection methods (Item 9), data items (Items 10a–10b), and study characteristics (Item 17). The synthesis phase is structured around effect measures (Item 12) and synthesis methods, data presentation, heterogeneity analysis, and sensitivity testing (Items 13a–13f), along with the results of individual studies (Item 19) and the results of syntheses (Items 20a–20d).

Quality assessment relates to the risk of bias in studies (Item 18), and the certainty of evidence (Item 22), which evaluates confidence in findings. Bias assessment is supported by the methods used to assess study risk of bias (Item 11), reporting bias assessment (Item 14), and bias due to missing results (Item 21) [57]. This alignment shows how PRISMA not only structures reporting but also reflects the operational phases of conducting a review.

The completion of the tasks is highly time-consuming. Rohit et al. conducted an analysis on the estimated time needed to complete 195 registered and completed reviews and found the mean time to be 67.3 weeks and the longest taking as long as 186 weeks [11]. These phases differ in duration and intensity, as many of the processes mentioned require much manual work such as, screening of studies, selection of studies and the data extraction. As these research studies encompass large parts or entire study domains, the amount of studies are usually in the thousands. Secondly, but perhaps most importantly this entire process must be reproducible by other researchers to maintain scientific rigor and ensure quality [92]. Moreover researches have to include all relevant studies, regardless of quality and minimize bias [92]. Ensuring quality in that scenario, extends the duration of the review processes. Given the extensive time investment, manual effort, and need for precision throughout the systematic review process, there is growing interest in leveraging technological solutions to support or automate key phases. In particular, Artificial Intelligence (AI) has emerged as a promising approach to enhance efficiency, reduce human error, and ensure reproducibility in evidence synthesis. The following section explores the field of AI and its potential contributions to the automation of systematic literature reviews.

1.3 Artificial Intelligence and Automation

Artificial Intelligence is a research domain that continues to be a subject of extensive debate and research. Especially surrounding the question "What is AI?". Throughout the years many different researchers have attempted to define the term *artificial intelligence*. John McCarthy, seen by some as the father of modern Artificial Intelligence research, was the first to attempt and define the term as "The science and engineering of making intelligent machines" [23]. Some decades later, Elain Rich defined AI as "The study of how to make computers do things that people are better at" [85]. This expanded the definition to include activities such as natural language understanding, problem solving and doing science [85]. While there is no widely accepted definition of AI, it is generally agreed upon that AI refers to a field of computer science dedicated to having systems perform task that usually require human intelligence. Essentially the integration of human-like intelligence into machines [47].

As the definition expanded to include broader range of activities, distinct subsets of Artificial Intelligence emerged, (1) Machine Learning, (2) Deep Learning, (3) Natural Language Processing and numerous subsets within these subsets. Machine Learning is a subset of AI and Deep Learning is a subset of ML, while Natural Language Processing is a subset of AI that crosses over into ML. Mirsha et al. [64] created a comprehensive visualization of the relationships between these concepts and subsequent subsets which can be seen in the figure below

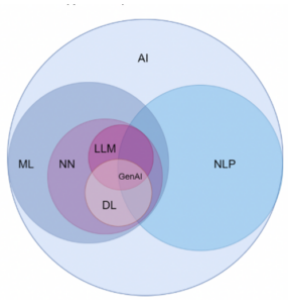

Fig. 1. Key Concepts of Artificial Intelligence and Machine Learning [64]

Machine learning encompasses the methods that machine(s) learn from using data. Using provided data - usually large data - a machine learns to make decisions without explicit programming [47]. It includes supervised learning, where the algorithm learns from labeled data to make predictions or classifications; unsupervised learning, which identifies patterns or structures in unlabeled data; and reinforcement learning, where an agent learns to make decisions by interacting with an environment and receiving feedback in the form of rewards or penalties.

Deep Learning expands on traditional Machine Learning by emulating the way our brain processes information. It accomplishes this through artificial neural networks [51]. It is therefore sometimes referred to as *deep structured learning* or *hierarchical training* [4].

Natural Language Processing (NLP) is another specialized area within AI concerned with the interaction between computers and human language. NLP methods are used to process, understand, and generate textual data, enabling applications like document summarization, sentiment analysis, and question answering [115].

Large Language Models (LLMs), such as BERT [22] and GPT [73], represent a recent advancement in NLP, developed using transformer-based architectures and trained on massive textual corpora. These models are capable of generating coherent text, understanding context, and performing downstream tasks such as classification and extraction with high accuracy [13]. LLMs leverage deep learning techniques and exemplify the convergence of AI, ML, DL, and NLP [63]. These subdomains are not mutually exclusive but rather hierarchically structured, with significant conceptual and technical overlap.

The continuous research and development of AI and it's subsets has opened up opportunities to automate several manual processes of SLRs and Meta-Analyses. There is potential for researches to save significant time and lower the entry barrier for new research. However despite the potential, this is still a relatively new area of application and a comprehensive overview of the current research is lacking. As a result, it is challenging to assess the current state of research and its future prospects.

1.4 Research Gap and Objective

Given the extensive demands of systematic reviews and the increasing interest in AI-assisted SLRs and/or Meta-Analyses, there is a need for a structured overview of existing AI tools that aid in the creation said reviews. Prior work has not comprehensively cataloged or critically assessed the tools currently available to support the core phases of systematic reviews and meta-analyses. This paper will analyze existing literature and conduct a systematic literature review to create a comprehensive overview of the existing tools for SLRs and Meta-Analyses. The aim is to answer the following research question:

What AI-based tools and frameworks exist to aid the key phases of Systematic Literature Reviews/Meta-Analyses and how do they contribute to the performance across key review phases?

To answer this question, the following sub-questions will be explored:

(1) RQ1: What AI-based tools are currently available for each of the core phases of SLRs and/or Meta-Analyses?

(2) RQ2: What type of AI technologies (e.g. ML, DL, NLP, LLM) are used across these tools and how do they align with the different phases?

(3) RQ3: Which performance metrics are most frequently used to evaluate AI tools in systematic reviews?

(4) RQ4: Which phases of the review process are underserved by the current AI tools?

2 METHODS

2.1 Information Sources

The studies on which this meta analyis was performed were acquired using 2 different methods. Firstly certain tools/papers were includes based on expert guidance. Secondly a Scopus search was performed on 15th February 2025.

2.1.1 Expert. Based on several meetings and e-mail exchanges several tools/papers were manually selected to be included for the analysis.

2.1.2 Scopus. The full search query can be seen in the appendix. As the full search query is quite long for readability purposes a simplified version of the search query can be seen below:

```
(TITLE-ABS-KEY("systematic review" OR "meta-analysis"
     OR "evidence synthesis"))
AND (TITLE-ABS-KEY("machine learning" OR "artificial
     intelligence" OR "LLM"
     OR "ChatGPT" OR "natural language processing" OR "
          automation"))
AND (TITLE-ABS-KEY("tool*" OR "software" OR "framework
     *" OR "screening"
     OR "data extraction" OR "study selection"))
AND NOT (TITLE-ABS-KEY("clinical" OR "patient*" OR "RCT
     " OR "hardware" OR "robot*"
     OR "student*" OR "pedagog*" OR "laboratory" OR "
          experiment*"
     OR "covid*" OR "cancer*" OR "infection*"))
```

This search query resulted in 6464 records. These search terms were chosen to capture papers discussing tools that assist in one or more processes involved in creating SLRs and Meta-Analyses.

2.2 Eligibility criteria

Based on the inclusion and exclusion criteria, studies will be accepted or rejected. The criteria are as follows:

- Study must evaluate or describe a tool
- Tool must be accessible to researchers
- Tool has to be for meta-analysis and/or systematic reviews
- Tool has to assist in at least one of the steps of the PRISMA 2020 framework
- Tool should include some form of AI (Artificial Intelligence) or ML (Machine Learning) methodologies. Based on the definitions provided

- Manuscript for tool must be written in English, Dutch or Bosnian (Serbo-Croatian); the author is proficient in these 3 languages.

These specific criteria were selected to acquire relevant tools and frameworks to answer the main RQ and exclude papers that include simply process description as that is not interesting for this research.

These specific criteria were selected to acquire relevant tools and frameworks to answer the main research question and to exclude papers that solely provide process descriptions, which are not of interest for this research. The screening process was conducted in two passes: first, papers were assessed based on their title and abstract; second, the full texts of the remaining papers were reviewed.

2.3 Search Strategy

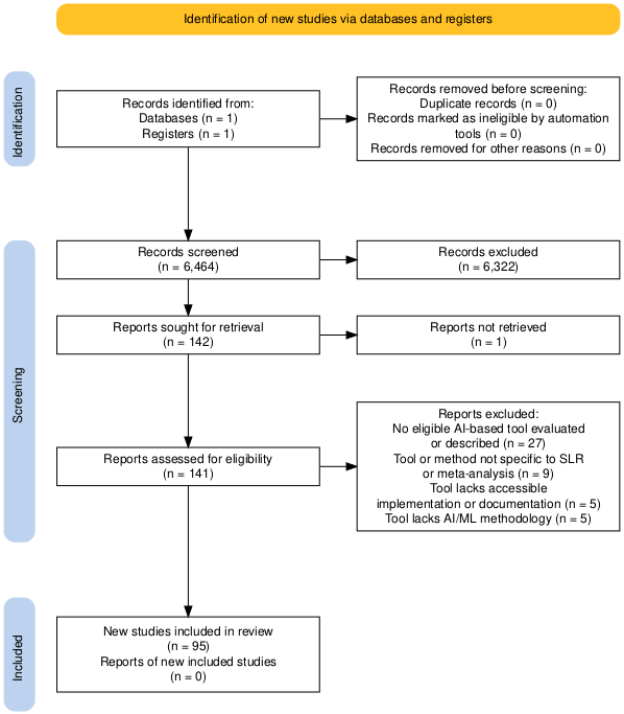

Fig. 2. Flow chart (Study Selection) [35]

2.4 Data Items

From the included papers, data variables were extracted as completely as possible.

Codebook Variables.

(1) **Tool Name**: Name of the tool or framework (e.g., ASReview)
(2) **References**: Citation identifiers of the study or studies discussing the tool (e.g., [30, 31])
(3) **Type**: Classification of the entry as a tool or framework (i.e., tool, framework)
(4) **PRISMA 2020 Phase Support**: PRISMA review phases supported by the tool (e.g., Identification, Screening, Data Extraction, etc.)
(5) **AI Type**: Specific AI technology implemented in the tool (e.g., ML, NLP, LLM, DL)
(6) **Performance Metrics**: Evaluation metrics reported for the tool (e.g., Accuracy, Sensitivity, Precision, Specificity, F1-score, AUC, Workload savings)

3 RESULTS

For readability purposes, a condensed and alphabetically ordered version of the extracted data is presented in table 1 below. This table provides an overview of the most relevant information and given that some tools were discussed across multiple studies, duplicate entries have been merged, with the primary reference listing all relevant studies. For those interested the full table can be seen in the Appendix.

PRISMA Phase Abbreviations and Performance Metrics

Abbrev.	PRISMA Phase	Performance Metrics (with Definitions)
I	Identification	–
S	Screening	• Sensitivity/Recall: Proportion of relevant documents correctly identified • Specificity: Proportion of irrelevant documents correctly excluded • Precision: Proportion of selected documents that are relevant • F1-score: Harmonic mean of precision and recall • WSS@R: Work Saved over Sampling at recall level R • AUC: Area Under the ROC Curve
DE	Data Extraction	• AccuracyDE: Proportion of correctly extracted data points • Completeness: Proportion of required data successfully extracted
DS	Data Synthesis	• Coherence Score: Measure of thematic consistency in synthesis • Similarity Score: Measure of conceptual alignment between studies
QA	Quality Assessment	• AgreementQA: Inter-rater reliability in quality judgments • AccuracyQA: Compared to expert human assessments
B	Bias Assessment	• AgreementB: Consistency with established bias evaluation frameworks • Detection Rate: Proportion of actual biases correctly identified

Table 1. Condensed Systematic Review Tools Overview (128 Tools) [20]

Tool Name	References	Type	I	S	DE	DS	QA	B	AI Type	Performance Metrics
2weekSR	[86, 91]	tool	✓	✓	✓				ML	-
Abstrackr	[10, 29–31, 60, 71, 80, 86, 89, 105]	tool		✓					ML	sensitivity[30, 31], specificity[30], precision[29, 30]
Active Pubmed Search (APS)	[89]	tool	✓						ML	-
ASReview	[9, 10, 31, 80, 81, 86]	tool		✓					ML	-
Atlas Ti	[71]	tool			✓				ML	-
Automatic NMA Screener	[55]	tool		✓					ML: NLP	sensitivity[55], workload decrease[55]
AutoLit	[40]	tool			✓				ML	-
BiblioShiny	[74]	tool	✓						ML: Text Mining	-
Carrot2	[74]	tool	✓						ML: Text Mining	-
ChatGPT (no version)	[34, 44, 48, 59, 109]	tool	✓	✓	✓				LLM	accuracyDE[44, 48], sensitivity[44]
ChatGPT3	[95]	framework		✓					LLM	sensitivity score[95]
ChatGPT3.5	[3, 21, 43, 56, 65, 68, 82, 88, 94]	tool	✓	✓					LLM	accuracy[56, 68, 88], specificity[56]
ChatGPT3.5-Turbo	[45]	tool		✓					LLM	precision[45], NPV[45]
ChatGPT4	[21, 34, 38, 54, 56, 59, 83, 97, 101]	tool	✓	✓	✓			✓	LLM	accuracyDE[34, 56], sensitivity[54, 56]
Chatpdf	[93]	tool			✓				ML	-
Citation Screener	[6, 52]	tool		✓					ML	precision[6, 52], recall[6]
Citation Screener (BioMedical)	[107]	tool		✓					ML	reduction manual screening[107]
CitNetExplorer	[74]	tool	✓						ML: Text Mining	-
Claude3	[56, 59, 83]	tool	✓	✓	✓				LLM	accuracyDE[56, 83]
Cochrane	[60, 89]	tool		✓					ML	-
Colandr	[10, 31, 86, 89]	tool	✓	✓					ML	-
Covidence	[31, 59, 71]	tool		✓					ML	-
Cruise Screening	[53]	tool		✓					ML	-
Data Extracter	[32]	tool			✓				ML	recall[32], precision[32]

Continued on next page

Table 1 – continued

Tool Name	References	Type	I	S	DE	DS	QA	B	AI Type	Performance Metrics
Deducklick	[86]	tool	✓						ML	-
Dextr	[10, 106]	tool			✓				ML	precision[106], recall[106]
DistillerSR	[10, 31, 59, 71, 86, 89]	tool	✓	✓	✓				ML	-
DocAnalyzer AI	[93]	tool		✓	✓				ML	-
DoCTER	[84]	tool		✓					ML	ensemble score[84]
Document Screener	[37]	tool		✓					ML	F1-score[37], accuracy[37]
Easy-peasy AI	[93]	tool							LLM	-
Elicit	[31, 46, 99, 100, 118]	tool	✓	✓	✓				LLM/ML	recall[118], precision[118]
EndNote	[74]	tool	✓						ML: Text Mining	-
EPPI-Reviewer	[10, 31, 59, 60, 71, 86, 89, 105]	tool	✓	✓		✓			ML	-
ExaCT	[10, 79, 86, 89]	tool			✓				ML: Text Mining	-
FAST2	[10]	tool		✓					ML	-
FastRead	[72]	tool							NLP	-
FSL-based screening	[113]	framework		✓					ML	workload reduction[113]
GAPScreener	[89]	tool		✓					ML	-
GATE	[86]	tool			✓				ML	-
Gemini 1.0	[56]	tool		✓					LLM	accuracy[56], specificity[56]
General-Purpose Filtering System	[49]	tool		✓					NLP	area under curve (AUC)[49]
Google PaLM2	[56]	tool		✓					LLM	accuracy[56], specificity[56]
GRADEpro	[71]	tool					✓		LLM	-
Hipdf	[93]	tool			✓				ML	-
IBM Parse	[89]	tool	✓						ML: Text Mining	-
In-Vitro Screener	[111]	tool		✓					ML: Text Mining	sensitivity[111], specificity[111]
Iris.ai	[10]	tool		✓	✓				ML	-
JBI-SUMARI	[71]	tool		✓	✓		✓		NLP	-
Keyword-Analyzer	[74]	tool	✓						ML: Text Mining	-
KeyCharacteristic Screener	[102]	tool		✓					ML: Text Mining	recall[102], exclusion[102]
Leximancer	[86]	tool	✓						ML	-
Literature Search Sandbox	[1]	tool	✓						LLM	sensitivity score[1]
LitSearchR	[86]	tool	✓						ML	-
LitSuggest	[10, 89]	tool	✓	✓					ML	-
LLama2	[56]	tool		✓					LLM	accuracy[56], specificity[56]
Llama3	[56]	tool		✓					LLM	accuracy[56], specificity[56]
LLM-FA	[17]	tool		✓					LLM	accuracy score[17]
LLM-SA	[17]	tool		✓					LLM	accuracy score[17]
Lingo3G	[74]	tool	✓						ML: Text Mining	-
Lingo4G	[74]	tool	✓						ML: Text Mining	-
MeSH on Demand	[71, 74]	tool	✓						ML: Text Mining	-
MetaCyto	[42]	tool			✓				ML	-
MetaMap	[86]	tool		✓					ML	-
MetaPreg	[89]	tool	✓	✓	✓				ML: NLP	-
ML-based screening	[18, 62, 77, 103]	tool	✓	✓					ML	specificity[62], ROC curves[62]
MMDas-AE	[89]	tool		✓					NLP	-
MySLR	[5]	tool	✓	✓	✓	✓			ML: Text Mining, NLP	-
NaCTeM	[60, 89]	tool			✓				ML: Text Mining	-
Natural Language Query's	[108]	tool	✓						NLP	recall[108], accuracy[108]
Nested Knowledge	[10, 31]	tool	✓	✓		✓			ML	-
NLP-based screening	[19, 67, 96, 116, 117]	tool	✓	✓	✓				NLP	recall[67], precision[67]
Nvivo	[71]	tool							ML	-
PDFBox	[86]	tool			✓				ML	-
Paperfetcher	[86]	tool	✓						ML	-
Pdf2gpt	[93]	tool			✓				ML	-
PICOPortal	[10]	tool		✓					ML	-

Continued on next page

Table 1 – continued

Tool Name	References	Type	I	S	DE	DS	QA	B	AI Type	Performance Metrics
Pitts.ai	[10]	tool		✓					ML	-
Polygot	[71]	tool							NLP	-
PublicationBias	[86]	tool						✓	ML	-
PubReMiner	[71, 74]	tool	✓						ML: Text Mining	-
Pythia	[2]	tool		✓					ML: NLP	precision[2], sensitivity[2]
RAPTOR: RevMan	[60]	tool			✓				ML	-
RapidMiner	[86]	tool		✓					ML	-
RCT Tagger	[60, 86]	tool	✓						ML	-
Rayyan	[10, 31, 71, 80, 86, 89, 105]	tool		✓					ML	-
ResearchRabbit	[46]	tool	✓	✓					NLP	-
ResearchScreener	[10, 16, 71, 72, 89]	tool		✓					ML	workload saving[16]
RevTools	[71, 89]	tool				✓			NLP: Text Mining	-
RobuMeta	[86]	tool						✓	ML	-
RobotAnalyst	[10, 60, 71, 72, 86, 89]	tool		✓					ML	-
RobotReviewer	[10, 31, 60, 86, 89]	tool		✓	✓		✓	✓	ML	-
RobotSearcher	[60, 86]	tool	✓						ML	-
SciSpace	[31, 93, 99, 100, 118]	tool	✓	✓	✓				ML/LLM	recall[118], precision[118]
SciSpaceGPT	[31]	tool			✓		✓		DL	-
Screen Smarter not Harder	[15]	tool		✓					ML	recall[15], sensitivity[15]
Screen4Me	[69, 89]	tool	✓	✓					ML	agreement[69], sensitivity[69]
SearchRefiner	[31]	tool	✓						ML	-
SLR Qub	[86]	tool	✓						ML	-
SLuRp	[12]	tool							ML	-
SMAFIRA	[14]	tool	✓						DL	-
SPARK	[7]	tool	✓	✓	✓				NLP (LDA)	coherence score[7]
srBERT	[8, 39]	tool		✓					ML: NLP	recall[8, 39], F1-score[8]
SRToolBox	[89]	tool	✓	✓	✓	✓			ML	-
SWARM-SLR	[112]	framework	✓	✓	✓	✓	✓	✓	AI	-
SWIFT-Active Screener	[10, 31, 58, 71, 89]	tool		✓					ML	agreement[58]
SWIFT-Review	[10, 31, 41, 60, 86, 89]	tool	✓	✓					ML	WSS@R[41]
SYMBALS	[69]	tool		✓					ML: Active Learning	speed (factor 6)[69]
SyRF	[86]	tool		✓					ML	-
SyROCCo	[26]	tool		✓	✓	✓			ML	similarity score[26]
SysRev	[10]	tool		✓					ML	-
Systematic Review Accelerator	[71]	tool		✓	✓				ML	-
TerMine	[74, 86]	tool	✓						ML: Text Mining	-
Text Analyzer	[74]	tool	✓						ML: Text Mining	-
Text Classifier	[28]	tool		✓					ML	precision[28], recall[28]
Thalia	[60]	tool	✓						ML	-
TIAB-Screener	[31, 78]	tool		✓					ML	area under curve[78]
Tm for R	[74]	tool	✓						ML: Text Mining	-
Transformer	[27]	tool		✓					ML	workload saved (WSS@R)[27]
Trial2Rev	[61]	tool	✓	✓	✓				ML	-
Twister	[50, 89]	tool	✓	✓					ML: Text Mining	-
VosViewer	[74]	tool	✓						ML: Text Mining	-
Voyant	[74]	tool	✓						ML: Text Mining	-
WordNet	[86]	tool	✓						ML	-
Yale MESH	[71, 74]	tool	✓						ML: Text Mining	-
Zettair	[86]	tool	✓						ML	-

3.1 Research Question 1

What AI-based tools are currently available for each of the core phases of SLRs and/or Meta-Analyses?

A total of **128 unique tools** were identified [1–3, 5–10, 12, 14–19, 21, 26–32, 34, 37–46, 48–50, 52–56, 58–62, 65, 67–69, 71, 72, 74, 77, 79–84, 86, 88, 89, 91, 93–97, 99–103, 105–109, 111–113, 116–118].

- **Identification (I)**: 55 tools (43.0%)
- **Screening (S)**: 72 tools (56.3%)
- **Data Extraction (DE)**: 34 tools (26.6%)
- **Data Synthesis (DS)**: 7 tools (5.5%)
- **Quality Assessment (QA)**: 5 tools (3.9%)
- **Bias Assessment (B)**: 5 tools (3.9%)

Close to a quarter (23.4%, n=30) of tools supported multiple phases with the most common combinations being Identification & Screening , Screening & Data Extraction, and Identification & Screening & Data Extraction.

3.2 Research Question 2

What type of AI technologies (e.g. ML, DL, NLP, LLM) are used across these tools and how do they align with the different phases?

- **Machine Learning (ML)** was the most common AI technique:
  - Used in 94 tools (73.4%).
  - Frequently implemented using methods like active learning and text mining.
- **Large Language Models (LLMs)** were used in 15 tools (11.7%):
  - Included general-purpose models (e.g., ChatGPT, Claude) and custom applications.
- **Natural Language Processing (NLP)** appeared in 12 tools (9.3%):
  - Often used as a component of ML pipelines.
- **Deep Learning (DL)** was applied in 2 tools (1.6%):
  - Typically overlapping with LLM or NLP-based functionality.

Across phases, ML-supported tools were dominant in screening, identification, and data extraction. LLMs primarily supported early phases such as screening and identification, with only limited coverage of later stages like quality or bias assessment. NLP-based tools focused mainly on screening and data extraction, while DL-based tools showed limited phase alignment and were mostly integrated in multi-method systems. Later phases such as data synthesis, quality assessment, and bias assessment were rarely supported by LLM, DL, or NLP tools. When supported, these phases were mostly addressed through machine learning models. A visualization of the distribution can be seen in table 2.

3.2.1 Statistical Analysis. A chi-square test of independence was conducted to assess whether the distribution of AI technology types (ML, NLP, LLM, DL) significantly varies across different systematic review phases. The test did not reveal a statistically significant association between AI technology type and review phase, $\chi^2(15, N = 189) = 17.48$, $p = 0.290$. This suggests that the overall usage of different AI technologies is not strongly dependent on the phase of the systematic review.

This statistical test was calculated using R Studio and the relevant code can be found in the appendix.

AI Technology	I	S	DE	DS	QA	B	Total
ML	45	53	26	6	2	5	137
NLP	5	10	4	1	1	0	21
LLMs	7	14	5	0	1	1	28
DL	1	0	1	0	1	0	3
Total	58	77	36	7	5	6	189

Table 2. Distribution of AI Technologies Across Systematic Review Phases

Note that totals do not match those of RQ 1, this is accurate due to the fact that tools can support one or multiple phases. Therefore tools that support multiple phases are counted once for each phase they cover in this particular table. Moreover certain tools use a combination of AI technologies.

3.3 Research Question 3

Which performance metrics are most frequently used to evaluate AI tools in systematic reviews?

Several performance metrics appear quite frequently. Out of the 128 tools surveyed, **43 tools (33.6%)** reported performance metrics. Among these, the most frequently reported metrics are **accuracy**, **sensitivity**, and **precision**.

- **Accuracy** is the most widely used metric, reported in 12 out of 43 tools (27.9%). It appears across a broad range of AI tools, including large language models (LLMs) and traditional machine learning classifiers. It measures the proportion of correct predictions among the total number of cases examined.
- **Sensitivity** (also known as recall) is reported in 14 out of 43 tools (32.6%). It reflects the tool's ability to correctly identify relevant studies or data points. This metric is particularly important in the early screening stages to ensure minimal exclusion of pertinent literature.
- **Precision** appears in 9 out of 43 tools (20.9%). It assesses the proportion of relevant instances among the retrieved ones and is frequently used in classification tasks, especially when balancing inclusion and exclusion criteria.

Other metrics, such as specificity, F1-score, negative predictive value (NPV), area under the ROC curve (AUC), and workload reduction, are also employed depending on the context and purpose of the AI tool. Finally, a few papers created

their own workload metrics separate from any standard metrics.

3.4 Research Question 4

Which phases of the review process are underserved by the current AI tools?

The later phases of the PRISMA-aligned review process are significantly underserved. Specifically:

- **Quality Assessment (QA)** was supported by only 5 tools (3.9%).
- **Bias Assessment (B)** was supported by 5 tools (3.9%).
- **Data Synthesis (DS)** was supported by 7 tools (5.5%).

In contrast, early phases showed much higher tool support:

- **Screening (S)**:72 tools (56.3%)
- **Identification (I)**: 55 tools (43.0%)
- **Data Extraction (DE)**: 34 tools (26.6%)

Only a single framework (SWARM-SLR) supported all six PRISMA phases. RobotReviewer was the only tool that bridged both early (screening, extraction) and late phases (quality and bias assessment). Most tools likely prioritize the early phases due to their manual repetitiveness and ease of automation relative to the complexity and subjectivity of later stages. This likely contributes to their lower rate of AI-based support.

4 DISCUSSION

128 unique AI-based tools were identified all designed to support one or multiple phases of the SLR/Meta-analysis process. The distribution of the tools across the PRISMA 2020 phases reveal a strong emphasis on screening (56.3%), followed by identification (43.0%) and data extraction (26.6%), while data synthesis (5.5%), quality assessment (3.9%) and bias assessment (3.9%) are underrepresented. These findings reflect a broader trend within the field: AI tools are primarily being developed to automate labor-intensive, repetitive processes that involve high document volumes and structured decision-making, such as initial filtering and eligibility screening.

The underrepresentation of AI support in later phases such as Data Synthesis, Quality Assessment, and Bias Assessment raises important implications. These phases often involve subjective judgment that remains difficult for current AI systems to replicate. As a result, methodological rigor and critical appraisal remain largely human-driven. This suggests a need for future tools not only to expand phase coverage but also to support human-AI collaboration through explainable and assistive models rather than fully automated replacements. Perhaps this should not be as surprising at all, considering the most time-consuming phases are the most manually labor-intensive ones, these include the same repetitive task over and over again and are easier to automate using artificial intelligence due to the lack of variables.

While some tools supported multiple phases (23.4%, n=30) of tools, it was still quite rare. Multi-phase tools most commonly had combinations of Identification & Screening & Data Extraction. Only a singular framework was identified that covered the entire process (SWARM-SLR) and the only tool covering multiple initial and later phases was RobotReviewer. This does show that tools like RobotReviewer, which attempt to bridge the gap between early and late phases are starting to emerge. But they remain the exception rather than the standard. Most tools still focus on isolated solutions for single phases.

A similar disparity was shown among the type of AI employed across the PRISMA phases. Machine Learning (ML) was the most frequently used technique, followed up by LLM, NLP and finally DL. ML, NLP and LLM were all predominately represented in the early phases (I, S, DE) and DL showed little to no phase alignment. It's occurrence was to little to make any relevant conclusion. However, A chi-square analysis confirmed that AI technology type was not significantly associated with review phase.

Performance evaluation practices also varied; numerous studies introduced performance metrics, in particular tools in the earlier phases, like those that covered identification, screening and data extraction. With accuracy, sensitivity (recall), and precision being the most frequently reported metrics. However, it is hard to compare even a tool covering the same phase using the same performance metrics, as tools may use different AI techniques, datasets or different versions of the same LLM. It is therefore necessary to create a standardized pipeline for testing tools across numerous phases of the SLR on numerous domains. For future work on top of an standardized benchmark (tool/framework), an open-source unified architecture that covers all phases should be introduced.

Finally the limitation of this study are as follows. While an attempt was made to make the search query as broadly as possible, it is still possible that the search may have missed relevant papers. Secondly the screening and data extraction was done by a single person and human's are prone to error and therefore the author could have missed relevant papers during the screening process or data extraction phase. Thirdly the review relied on the reported metrics of the papers and conducted no hands on testing, which may have led to overestimating or underestimating certain tools and many tools did not report any key metrics leading to a gap in performance data. Validation is necessary and should be the next step.

REFERENCES

[1] Gaelen P Adam, Jay DeYoung, Alice Paul, Ian J Saldanha, Ethan M Balk, Thomas A Trikalinos, and Byron C Wallace. 2024. Literature search sandbox: a large language model that generates search queries for systematic reviews. *JAMIA Open* 7, 3 (July 2024). https://doi.org/10.1093/jamiaopen/ooae098

[2] Gaelen P. Adam, Dimitris Pappas, Haris Papageorgiou, Evangelos Evangelou, and Thomas A. Trikalinos. 2022. A novel tool that allows interactive screening of PubMed citations showed promise for the semi-automation of identification of Biomedical Literature. *Journal of Clinical Epidemiology* 150 (Oct. 2022), 63–71. https://doi.org/10.1016/j.jclinepi.2022.06.007

[3] Ahmad Alshami, Moustafa Elsayed, Eslam Ali, Abdelrahman E. E. Eltoukhy, and Tarek Zayed. 2023. Harnessing the Power of ChatGPT for Automating Systematic Review Process: Methodology, Case Study, Limitations, and Future Directions. *Systems* 11, 7 (July 2023), 351. https://doi.org/10.3390/systems11070351

[4] Thimira Amaratunga and Thimira Amaratunga. 2021. What is deep learning? *Deep Learning on Windows: Building Deep Learning Computer*

Vision Systems on Microsoft Windows (2021), 1–14.

[5] Salvatore Ammirato, Alberto M. Felicetti, Daniele Rogano, Roberto Linzalone, and Vincenzo Corvello. 2022. Digitalising the Systematic Literature Review process: the MySLR platform. *Knowledge Management Research &; Practice* 21, 4 (Feb. 2022), 777–794. https://doi.org/10.1080/14778238.2022.2041375

[6] Dhrangadhariya Anjani, Hilfiker Roger, Schaer Roger, and Müller Henning. 2020. *Machine Learning Assisted Citation Screening for Systematic Reviews*. IOS Press. https://doi.org/10.3233/shti200171

[7] Cameron Frederick Atkinson. 2025. AI-pocalypse now: Automating the systematic literature review with SPARK (Systematic processing and automated review Kit) – gathering, organising, filtering, and scaffolding. *MethodsX* 14 (June 2025), 103129. https://doi.org/10.1016/j.mex.2024.103129

[8] Sungmin Aum and Seon Choe. 2021. srBERT: automatic article classification model for systematic review using BERT. *Systematic Reviews* 10, 1 (Oct. 2021). https://doi.org/10.1186/s13643-021-01763-w

[9] Rodrigo Cupertino Bernardes, Lorena Lisbetd Botina, Renan dos Santos Araújo, Raul Narciso Carvalho Guedes, Gustavo Ferreira Martins, and Maria Augusta Pereira Lima. 2022. Artificial Intelligence-Aided Meta-Analysis of Toxicological Assessment of Agrochemicals in Bees. *Frontiers in Ecology and Evolution* 10 (May 2022). https://doi.org/10.3389/fevo.2022.845608

[10] Francisco Bolaños, Angelo Salatino, Francesco Osborne, and Enrico Motta. 2024. Artificial intelligence for literature reviews: opportunities and challenges. *Artificial Intelligence Review* 57, 10 (Aug. 2024). https://doi.org/10.1007/s10462-024-10902-3

[11] Rohit Borah, Andrew W Brown, Patrice L Capers, and Kathryn A Kaiser. 2017. Analysis of the time and workers needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. *BMJ Open* 7, 2 (Feb. 2017), e012545. https://doi.org/10.1136/bmjopen-2016-012545

[12] David Bowes, Tracy Hall, and Sarah Beecham. 2012. SLuRp: a tool to help large complex systematic literature reviews deliver valid and rigorous results. In *Proceedings of the 2nd international workshop on Evidential assessment of software technologies (ESEM '12)*. ACM, 33–36. https://doi.org/10.1145/2372233.2372243

[13] Tom B Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in Neural Information Processing Systems* 33 (2020), 1877–1901.

[14] Daniel Butzke, Bettina Bert, Konrad Gulich, Gilbert Schönfelder, and Mariana Neves. 2024. SMAFIRA: a literature-based web tool to assist researchers with retrieval of 3R-relevant information. *Laboratory Animals* (June 2024). https://doi.org/10.1177/00236772241237608

[15] Diego G. Campos, Tim Fütterer, Thomas Gfrörer, Rosa Lavelle-Hill, Kou Murayama, Lars König, Martin Hecht, Steffen Zitzmann, and Ronny Scherer. 2024. Screening Smarter, Not Harder: A Comparative Analysis of Machine Learning Screening Algorithms and Heuristic Stopping Criteria for Systematic Reviews in Educational Research. *Educational Psychology Review* 36, 1 (Feb. 2024). https://doi.org/10.1007/s10648-024-09862-5

[16] Kevin E. K. Chai, Robin L. J. Lines, Daniel F. Gucciardi, and Leo Ng. 2021. Research Screener: a machine learning tool to semi-automate abstract screening for systematic reviews. *Systematic Reviews* 10, 1 (April 2021). https://doi.org/10.1186/s13643-021-01635-3

[17] Haichao Chen, Zehua Jiang, Xinyu Liu, Can Can Xue, Samantha Min Er Yew, Bin Sheng, Ying-Feng Zheng, Xiaofei Wang, You Wu, Sobha Sivaprasad, Tien Yin Wong, Varun Chaudhary, and Yih Chung Tham. 2025. Can large language models fully automate or partially assist paper selection in systematic reviews? *British Journal of Ophthalmology* (Jan. 2025), bjo–2024–326254. https://doi.org/10.1136/bjo-2024-326254

[18] Olga Chernikova, Matthias Stadler, Ivan Melev, and Frank Fischer. 2024. Using machine learning for continuous updating of meta-analysis in educational context. *Computers in Human Behavior* 156 (July 2024), 108215. https://doi.org/10.1016/j.chb.2024.108215

[19] Itzik David and Roy Gelbard. 2024. Using Machine Learning for Systematic Literature Review Case in Point: Agile Software Development. *WIREs Data Mining and Knowledge Discovery* 15, 1 (Oct. 2024). https://doi.org/10.1002/widm.1569

[20] DeepSeek. 2024. DeepSeek Coding. https://deepseek.com/coding. Assisted in creating LaTeX tables, accessed April 2025.

[21] Gizem Boztaş Demir, Yağızalp Süküt, Gökhan Serhat Duran, Kübra Gülnur Topsakal, and Serkan Görgülü. 2024. Enhancing systematic reviews in orthodontics: a comparative examination of GPT-3.5 and GPT-4 for generating PICO-based queries with tailored prompts and configurations. *European Journal of Orthodontics* 46, 2 (March 2024). https://doi.org/10.1093/ejo/cjae011

[22] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *arXiv preprint arXiv:1810.04805* (2019).

[23] Wolfgang Ertel. 2024. *Introduction to artificial intelligence*. Springer Nature.

[24] Tullio Facchinetti, Guido Benetti, Davide Giuffrida, and Antonino Nocera. 2022. slr-kit: A semi-supervised machine learning framework for systematic literature reviews. *Knowledge-Based Systems* 251 (Sept. 2022), 109266. https://doi.org/10.1016/j.knosys.2022.109266

[25] Robert H. Fagard, Jan A. Staessen, and Lutgarde Thijs. 1996. Advantages and disadvantages of the meta-analysis approach. *Journal of Hypertension* 14, Supplement 2 (Sept. 1996), S9–S13. https://doi.org/10.1097/00004872-199609002-00004

[26] Zheng Fang, Miguel Arana-Catania, Felix-Anselm van Lier, Juliana Outes Velarde, Harry Bregazzi, Mara Airoldi, Eleanor Carter, and Rob Procter. 2024. SyROCCo: enhancing systematic reviews using machine learning. *Data & Policy* 6 (2024). https://doi.org/10.1017/dap.2024.33

[27] Allan Victor Almeida Faria, Maísa Kely de Melo, Flávio Augusto R. de Oliveira, Li Weigang, and Victor Rafael Rezende Celestino. 2023. *Automated SLR with a Few Labeled Papers and a Fair Workload Metric*. Springer Nature Switzerland, 1–23. https://doi.org/10.1007/978-3-031-43088-6_1

[28] J.J. García Adeva, J.M. Pikatza Atxa, M. Ubeda Carrillo, and E. Ansuategi Zengotitabengoa. 2014. Automatic text classification to support systematic reviews in medicine. *Expert Systems with Applications* 41, 4 (March 2014), 1498–1508. https://doi.org/10.1016/j.eswa.2013.08.047

[29] Allison Gates, Michelle Gates, Meghan Sebastianski, Samantha Guitard, Sarah A. Elliott, and Lisa Hartling. 2020. The semi-automation of title and abstract screening: a retrospective exploration of ways to leverage Abstrackr's relevance predictions in systematic and rapid reviews. *BMC Medical Research Methodology* 20, 1 (June 2020). https://doi.org/10.1186/s12874-020-01031-w

[30] Allison Gates, Cydney Johnson, and Lisa Hartling. 2018. Technology-assisted title and abstract screening for systematic reviews: a retrospective evaluation of the Abstrackr machine learning tool. *Systematic Reviews* 7, 1 (March 2018). https://doi.org/10.1186/s13643-018-0707-8

[31] Lixia Ge, Rupesh Agrawal, Maxwell Singer, Palvannan Kannapiran, Joseph Antonio De Castro Molina, Kiok Liang Teow, Chun Wei Yap, and John Arputhan Abisheganaden. 2024. Leveraging artificial intelligence to enhance systematic reviews in health research: advanced tools and challenges. *Systematic Reviews* 13, 1 (Oct. 2024). https://doi.org/10.1186/s13643-024-02682-2

[32] Shubhaditya Goswami, Sukanya Pal, Simon Goldsworthy, and Tanmay Basu. 2019. *An Effective Machine Learning Framework for Data Elements Extraction from the Literature of Anxiety Outcome Measures to Build Systematic Review*. Springer International Publishing, 247–258. https://doi.org/10.1007/978-3-030-20485-3_19

[33] Dhruv Grewal, Nancy Puccinelli, and Kent B Monroe. 2018. Meta-analysis: integrating accumulated knowledge. *Journal of the Academy of Marketing Science* 46 (2018), 9–30.

[34] Eddie Guo, Mehul Gupta, Jiawen Deng, Ye-Jean Park, Michael Paget, and Christopher Naugler. 2024. Automated Paper Screening for Clinical Reviews Using Large Language Models: Data Analysis Study. *Journal of Medical Internet Research* 26 (Jan. 2024), e48996. https://doi.org/10.2196/48996

[35] Neal R Haddaway, Matthew J Page, Chris C Pritchard, and Luke A McGuinness. 2022. PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis. *Campbell Syst. Rev.* 18, 2 (June 2022), e1230.

[36] Binglan Han, Teo Susnjak, and Anuradha Mathrani. 2024. Automating Systematic Literature Reviews with Retrieval-Augmented Generation: A Comprehensive Overview. *Applied Sciences* 14, 19 (Oct. 2024), 9103. https://doi.org/10.3390/app14199103

[37] Abdelhakim Hannousse and Salima Yahiouche. 2022. *A Semi-automatic Document Screening System for Computer Science Systematic Reviews*. Springer International Publishing, 201–215. https://doi.org/10.1007/978-3-031-04112-9_15

[38] Bashar Hasan, Samer Saadi, Noora S Rajjoub, Moustafa Hegazi, Mohammad Al-Kordi, Farah Fleti, Magdoleen Farah, Irbaz B Riaz, Imon Banerjee, Zhen Wang, and Mohammad Hassan Murad. 2024. Integrating large language models in systematic reviews: a framework and case study using ROBINS-I for risk of bias assessment. *BMJ Evidence-Based Medicine* 29, 6 (Feb. 2024), 394–398. https://doi.org/10.1136/bmjebm-2023-112597

[39] Marta Hasny, Alexandru-Petru Vasile, Mario Gianni, Alexandra Bannach-Brown, Mona Nasser, Murray Mackay, Diana Donovan, Jernej Šorli, Ioana Domocos, Milad Dulloo, Nimita Patel, Olivia Drayson, Nicole Meerah Elango, Jéromine Vacquie, Ana Patricia Ayala, and Anna Fogtman. 2023. *BERT for Complex Systematic Review Screening to Support the Future of Medical Research*. Springer Nature Switzerland, 173–182. https://doi.org/10.1007/978-3-031-34344-5_21

[40] Karl Holub, Nicole Hardy, and Kevin Kallmes. 2021. Toward Automated Data Extraction According to Tabular Data Structure: Cross-sectional Pilot Survey of the Comparative Clinical Literature. *JMIR Formative Research* 5, 11 (Nov. 2021), e33124. https://doi.org/10.2196/33124

[41] Brian E. Howard, Jason Phillips, Kyle Miller, Arpit Tandon, Deepak Mav, Mihir R. Shah, Stephanie Holmgren, Katherine E. Pelch, Vickie Walker, Andrew A. Rooney, Malcolm Macleod, Ruchir R. Shah, and Kristina Thayer. 2016. SWIFT-Review: a text-mining workbench for systematic review. *Systematic Reviews* 5, 1 (May 2016). https://doi.org/10.1186/s13643-016-0263-z

[42] Zicheng Hu, Chethan Jujjavarapu, Jacob J. Hughey, Sandra Andorf, Hao-Chih Lee, Pier Federico Gherardini, Matthew H. Spitzer, Cristel G. Thomas, John Campbell, Patrick Dunn, Jeff Wiser, Brian A. Kidd, Joel T. Dudley, Garry P. Nolan, Sanchita Bhattacharya, and Atul J. Butte. 2018. MetaCyto: A Tool for Automated Meta-analysis of Mass and Flow Cytometry Data. *Cell Reports* 24, 5 (July 2018), 1377–1388. https://doi.org/10.1016/j.celrep.2018.07.003

[43] Aleksi Huotala, Miikka Kuutila, Paul Ralph, and Mika Mäntylä. 2024. The Promise and Challenges of Using LLMs to Accelerate the Screening Process of Systematic Reviews. In *Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering (EASE 2024)*. ACM, 262–271. https://doi.org/10.1145/3661167.3661172

[44] Mahbod Issaiy, Hossein Ghanaati, Shahriar Kolahi, Madjid Shakiba, Amir Hossein Jalali, Diana Zarei, Sina Kazemian, Mahsa Alborzi Avanaki, and Kavous Firouznia. 2024. Methodological insights into ChatGPT's screening performance in systematic reviews. *BMC Medical Research Methodology* 24, 1 (March 2024). https://doi.org/10.1186/s12874-024-02203-8

[45] Mahbod Issaiy, Hossein Ghanaati, Shahriar Kolahi, Madjid Shakiba, Amir Hossein Jalali, Diana Zarei, Sina Kazemian, Mahsa Alborzi Avanaki, and Kavous Firouznia. 2024. Methodological insights into ChatGPT's screening performance in systematic reviews. *BMC Medical Research Methodology* 24, 1 (March 2024). https://doi.org/10.1186/s12874-024-02203-8

[46] Seyed Mohammad Ali Jafari. 2024. Streamlining the Selection Phase of Systematic Literature Reviews (SLRs) Using AI-Enabled GPT-4 Assistant API. https://doi.org/10.48550/ARXIV.2402.18582

[47] D. Jakhar and I. Kaur. 2019. Artificial intelligence, machine learning and deep learning: definitions and differences. *Clinical and Experimental Dermatology* 45, 1 (July 2019), 131–132. https://doi.org/10.1111/ced.14029

[48] Chuanzhi Jiang, Sen Li, Di Cai, Jin Ye, Qinghang Bao, Cuiling Liu, and Songxue Wang. 2024. ChatGPT-based meta-analysis for evaluating the temporal and spatial characteristics of deoxynivalenol contamination in Chinese wheat. *Journal of Hazardous Materials* 480 (Dec. 2024), 135888. https://doi.org/10.1016/j.jhazmat.2024.135888

[49] Hawkins John and Tivey David. 2024. Literature Filtering for Systematic Reviews with Transformers. In *2024 2nd International Conference on Communications, Computing and Artificial Intelligence (CCCAI 2024)*. ACM, 1–7. https://doi.org/10.1145/3676581.3676582

[50] Karl Kreiner, Dieter Hayn, and Günter Schreier. 2018. Twister: A Tool for Reducing Screening Time in Systematic Literature Reviews. *Studies in health technology and informatics* 255 (01 2018), 5–9.

[51] Jakub Kufel, Katarzyna Bargieł-Łączek, Szymon Kocot, Maciej Koźlik, Wiktoria Bartnikowska, Michał Janik, Łukasz Czogalik, Piotr Dudek, Mikołaj Magiera, Anna Lis, Iga Paszkiewicz, Zbigniew Nawrat, Maciej Cebula, and Katarzyna Gruszczyńska. 2023. What Is Machine Learning, Artificial Neural Networks and Deep Learning?—Examples of Practical Applications in Medicine. *Diagnostics* 13, 15 (Aug. 2023), 2582. https://doi.org/10.3390/diagnostics13152582

[52] Wojciech Kusa, Allan Hanbury, and Petr Knoth. 2022. Automation of Citation Screening for Systematic Literature Reviews using Neural Networks: A Replicability Study. https://doi.org/10.48550/ARXIV.2201.07534

[53] Wojciech Kusa, Petr Knoth, and Allan Hanbury. 2023. CRUISE-Screening: Living Literature Reviews Toolbox. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM '23)*. ACM, 5071–5075. https://doi.org/10.1145/3583780.3614736

[54] Assaf Landschaft, Dario Antweiler, Sina Mackay, Sabine Kugler, Stefan Rüping, Stefan Wrobel, Timm Höres, and Hector Allende-Cid. 2024. Implementation and evaluation of an additional GPT-4-based reviewer in PRISMA-based medical systematic literature reviews. *International Journal of Medical Informatics* 189 (Sept. 2024), 105531. https://doi.org/10.1016/j.ijmedinf.2024.105531

[55] Ivan Lerner, Perrine Créquit, Philippe Ravaud, and Ignacio Atal. 2019. Automatic screening using word embeddings achieved high sensitivity and workload reduction for updating living network meta-analyses. *Journal of Clinical Epidemiology* 108 (April 2019), 86–94. https://doi.org/10.1016/j.jclinepi.2018.12.001

[56] Michael Li, Jianping Sun, and Xianming Tan. 2024. Evaluating the effectiveness of large language models in abstract screening: a comparative analysis. *Systematic Reviews* 13, 1 (Aug. 2024). https://doi.org/10.1186/s13643-024-02609-x

[57] A Liberati, DG Altman, J Tetzlaff, C Mulrow, PC Gotzsche, JPA Ioannidis, M Clarke, PJ Devereaux, J Kleijnen, and D Moher. 2009. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: Explanation and elaboration. BMJ, 339 (jul21 1), b2700–b2700.

[58] Jenny J. W. Liu, Natalie Ein, Julia Gervasio, Bethany Easterbrook, Maede S. Nouri, Anthony Nazarov, and J. Don Richardson. 2024. Usability and agreement of the SWIFT-ActiveScreener systematic review support tool: Preliminary evaluation for use in clinical research. *PLOS ONE* 19, 11 (Nov. 2024), e0291163. https://doi.org/10.1371/journal.pone.0291163

[59] Xufei Luo, Fengxian Chen, Di Zhu, Ling Wang, Zijun Wang, Hui Liu, Meng Lyu, Ye Wang, Qi Wang, and Yaolong Chen. 2024. Potential Roles of Large Language Models in the Production of Systematic Reviews and Meta-Analyses. *Journal of Medical Internet Research* 26 (June 2024), e56780. https://doi.org/10.2196/56780

[60] Iain J. Marshall and Byron C. Wallace. 2019. Toward systematic review automation: a practical guide to using machine learning tools in research synthesis. *Systematic Reviews* 8, 1 (July 2019). https://doi.org/10.1186/s13643-019-1074-9

[61] Paige Martin, Didi Surian, Rabia Bashir, Florence T Bourgeois, and Adam G Dunn. 2019. Trial2rev: Combining machine learning and crowdsourcing to create a shared space for updating systematic reviews. *JAMIA Open* 2, 1 (Jan. 2019), 15–22. https://doi.org/10.1093/jamiaopen/ooy062

[62] H.S. Menold, V.L.S Wieland, C.M. Haney, D. Uysal, F. Wessels, G.C. Cacciamani, M.S. Michel, S. Seide, and K.F. Kowalewski. 2024. Machine learning enables automated screening for systematic reviews and meta-analysis in urology. *World Journal of Urology* 42, 1 (July 2024). https://doi.org/10.1007/s00345-024-05078-y

[63] Shervin Minaee, Tomas Mikolov, Niloofar Nikzad, Mahdi Chenaghlu, Richard Socher, Xavier Amatriain, and Jianfeng Gao. 2024. Large Language Models: A Survey. *arXiv preprint arXiv:2401.06247* (2024). https://arxiv.org/abs/2401.06247

[64] Akanksha Mishra. 2024. A comprehensive review of Artificial Intelligence and Machine Learning : Concepts, trends, and applications. *Int J Sci Res Sci & Technol* 11, 5 (Sept. 2024), 126–142.

[65] Mehrnaz Mostafapour, Jacqueline H Fortier, Karen Pacheco, Heather Murray, and Gary Garber. 2024. Evaluating Literature Reviews Conducted by Humans Versus ChatGPT: Comparative Study. *JMIR AI* 3 (Aug. 2024), e56537. https://doi.org/10.2196/56537

[66] Mette Motzfeldt Jensen, Mathias Brix Danielsen, Johannes Riis, Karoline Assifuah Kristjansen, Stig Andersen, Yoshiro Okubo, and Martin Grønbech Jørgensen. 2025. ChatGPT-4o can serve as the second rater for data extraction in systematic reviews. *PLOS ONE* 20, 1 (Jan. 2025), e0313401. https://doi.org/10.1371/journal.pone.0313401

[67] Bianca Minetto Napoleão, Ritika Sarkar, Sylvain Hallé, Fabio Petrillo, and Marcos Kalinowski. 2024. Emerging Results on Automated Support for Searching and Selecting Evidence for Systematic Literature Review Updates. https://doi.org/10.48550/ARXIV.2402.05317

[68] Isabel Nicholson Thomas, Philip Roche, and Adrienne Grêt-Regamey. 2024. Harnessing artificial intelligence for efficient systematic reviews: A case study in ecosystem condition indicators. *Ecological Informatics* 83 (Nov. 2024), 102819. https://doi.org/10.1016/j.ecoinf.2024.102819

[69] Anna Noel-Storr, Gordon Dooley, Lisa Affengruber, and Gerald Gartlehner. 2021. Citation screening using crowdsourcing and machine learning produced accurate results: Evaluation of Cochrane's modified Screen4Me service. *Journal of Clinical Epidemiology* 130 (Feb. 2021), 23–31. https://doi.org/10.1016/j.jclinepi.2020.09.024

[70] Takehiko Oami, Yohei Okada, and Taka-aki Nakada. 2024. Performance of a Large Language Model in Screening Citations. *JAMA Network Open* 7, 7 (July 2024), e2420496. https://doi.org/10.1001/jamanetworkopen.2024.

20496

[71] Regina Ofori-Boateng, Magaly Aceves-Martins, Nirmalie Wiratunga, and Carlos Francisco Moreno-Garcia. 2024. Towards the automation of systematic reviews using natural language processing, machine learning, and deep learning: a comprehensive review. *Artificial Intelligence Review* 57, 8 (July 2024). https://doi.org/10.1007/s10462-024-10844-w

[72] Regina Ofori-Boateng, Tamy Goretty Trujillo-Escobar, Magaly Aceves-Martins, Nirmalie Wiratunga, and Carlos Francisco Moreno-Garcia. 2024. Enhancing systematic reviews: An in-depth analysis on the impact of active learning parameter combinations for biomedical abstract screening. *Artificial Intelligence in Medicine* 157 (Nov. 2024), 102989. https://doi.org/10.1016/j.artmed.2024.102989

[73] OpenAI. 2024. ChatGPT. https://chat.openai.com. Image generated using ChatGPT, accessed April 2025.

[74] Hannah O'Keefe, Judith Rankin, Sheila A. Wallace, and Fiona Beyer. 2022. Investigation of text-mining methodologies to aid the construction of search strategies in systematic reviews of diagnostic test accuracy—a case study. *Research Synthesis Methods* 14, 1 (July 2022), 79–98. https://doi.org/10.1002/jrsm.1593

[75] Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *bmj* 372 (2021).

[76] Justin Paul and Mojtaba Barari. 2022. Meta-analysis and traditional systematic literature reviews—What, why, when, where, and how? *Psychology & Marketing* 39, 6 (2022), 1099–1115.

[77] Bart G. Pijls. 2024. Machine Learning assisted systematic reviewing in orthopaedics. *Journal of Orthopaedics* 48 (Feb. 2024), 103–106. https://doi.org/10.1016/j.jor.2023.11.051

[78] Maximilian Pilz, Samuel Zimmermann, Juliane Friedrichs, Enrica Wördehoff, Ulrich Ronellenfitsch, Meinhard Kieser, and Johannes A. Vey. 2024. Semi-automated title-abstract screening using natural language processing and machine learning. *Systematic Reviews* 13, 1 (Nov. 2024). https://doi.org/10.1186/s13643-024-02688-w

[79] Richeek Pradhan, David C. Hoaglin, Matthew Cornell, Weisong Liu, Victoria Wang, and Hong Yu. 2019. Automatic extraction of quantitative data from ClinicalTrials.gov to conduct meta-analyses. *Journal of Clinical Epidemiology* 105 (Jan. 2019), 92–100. https://doi.org/10.1016/j.jclinepi.2018.08.023

[80] Yazhuo Quan, Tetiana Tytko, and Bronson Hui. 2024. Utilizing ASReview in screening primary studies for meta-research in SLA: A step-by-step tutorial. *Research Methods in Applied Linguistics* 3, 1 (April 2024), 100101. https://doi.org/10.1016/j.rmal.2024.100101

[81] Yazhuo Quan, Tetiana Tytko, and Bronson Hui. 2024. Utilizing ASReview in screening primary studies for meta-research in SLA: A step-by-step tutorial. *Research Methods in Applied Linguistics* 3, 1 (April 2024), 100101. https://doi.org/10.1016/j.rmal.2024.100101

[82] Riaz Qureshi, Daniel Shaughnessy, Kayden A. R. Gill, Karen A. Robinson, Tianjing Li, and Eitan Agai. 2023. Are ChatGPT and large language models "the answer" to bringing us closer to systematic review automation? *Systematic Reviews* 12, 1 (April 2023). https://doi.org/10.1186/s13643-023-02243-z

[83] Tim Reason, Emma Benbow, Julia Langham, Andy Gimblett, Sven L. Klijn, and Bill Malcolm. 2024. Artificial Intelligence to Automate Network Meta-Analyses: Four Case Studies to Evaluate the Potential Application of Large Language Models. *PharmacoEconomics - Open* 8, 2 (Feb. 2024), 205–220. https://doi.org/10.1007/s41669-024-00476-9

[84] Ashleigh Rich, Emma McGorray, Carrie Baldwin-SoRelle, Michelle Cawley, Karen Grigg, Lauren B. Beach, Gregory Phillips, and Tonia Poteat. 2025. Automated tools for systematic review screening methods: an application of machine learning for sexual orientation and gender identity measurement in health research. *Journal of the Medical Library Association* 113, 1 (Jan. 2025), 31–38. https://doi.org/10.5195/jmla.2025.1860

[85] Elaine Rich. 1985. Artificial intelligence and the humanities. *Computers and the Humanities* 19, 2 (1985), 117–122.

[86] Stephanie Roth and Alex Wermer-Colan. 2023. Machine learning methods for systematic reviews:: A rapid scoping review. *Dela. J. Public Health* 9, 4 (Nov. 2023), 40–47.

[87] Chris Rowley and Justin Paul. 2021. Introduction: the role and relevance of literature reviews and research in the Asia Pacific. *Asia Pacific Business Review* 27 (04 2021), 1–5. https://doi.org/10.1080/13602381.2021.1894839

[88] Suebsarn Ruksakulpiwat, Lalipat Phianhasin, Chitchanok Benjasirisan, Kedong Ding, Anuoluwapo Ajibade, Ayanesh Kumar, and Cassie Stewart. 2024. Assessing the Efficacy of ChatGPT Versus Human Researchers in Identifying Relevant Studies on mHealth Interventions for Improving Medication Adherence in Patients With Ischemic Stroke When Conducting Systematic Reviews: Comparative Analysis. *JMIR mHealth and uHealth* 12 (May 2024), e51526. https://doi.org/10.2196/51526

[89] Álisson Oliveira dos Santos, Eduardo Sergio da Silva, Letícia Machado Couto, Gustavo Valadares Labanca Reis, and Vinícius Silva Belo. 2023. The use of artificial intelligence for automating or semi-automating biomedical literature analyses: A scoping review. *Journal of Biomedical Informatics* 142 (June 2023), 104389. https://doi.org/10.1016/j.jbi.2023.104389

[90] Lena Schmidt, Farhad Shokraneh, Kirsten Steinhausen, and Clive E. Adams. 2019. Introducing RAPTOR: RevMan Parsing Tool for Reviewers. *Systematic Reviews* 8, 1 (June 2019). https://doi.org/10.1186/s13643-019-1070-0

[91] Anna Mae Scott, Paul Glasziou, and Justin Clark. 2023. We extended the 2-week systematic review (2weekSR) methodology to larger, more complex systematic reviews: A case series. *Journal of Clinical Epidemiology* 157 (May 2023), 112–119. https://doi.org/10.1016/j.jclinepi.2023.03.007

[92] Andy P. Siddaway, Alex M. Wood, and Larry V. Hedges. 2019. How to Do a Systematic Review: A Best Practice Guide for Conducting and Reporting Narrative Reviews, Meta-Analyses, and Meta-Syntheses. *Annual Review of Psychology* 70, 1 (Jan. 2019), 747–770. https://doi.org/10.1146/annurev-psych-010418-102803

[93] Lotfi Souifi, Nesrine Khabou, Ismael Rodriguez, and Ahmed Kacem. 2024. Towards the Use of AI-Based Tools for Systematic Literature Review. In *Proceedings of the 16th International Conference on Agents and Artificial Intelligence*. SCITEPRESS - Science and Technology Publications, 595–603. https://doi.org/10.5220/0012467700003636

[94] Scott Spillias, Paris Tuohy, Matthew Andreotta, Ruby Annand-Jones, Fabio Boschetti, Christopher Cvitanovic, Joseph Duggan, Elisabeth A. Fulton, Denis B. Karcher, Cécile Paris, Rebecca Shellock, and Rowan Trebilco. 2024. Human-AI collaboration to identify literature for evidence synthesis. *Cell Reports Sustainability* 1, 7 (July 2024), 100132. https://doi.org/10.1016/j.crsus.2024.100132

[95] James A Strachan. 2025. Designing GPT3 prompts to screen articles for systematic reviews of RCTs. *Health Policy and Technology* 14, 1 (Feb. 2025), 100943. https://doi.org/10.1016/j.hlpt.2024.100943

[96] Girish Sundaram and Daniel Berleant. 2023. *Automating Systematic Literature Reviews with Natural Language Processing and Text Mining: A Systematic Literature Review*. Springer Nature Singapore, 73–92. https://doi.org/10.1007/978-981-99-3243-6_7

[97] Lefteris TEPERIKIDIS, Aristi BOULMPOU, Christodoulos PAPADOPOULOS, and Giuseppe BIONDI-ZOCCAI. 2024. Using ChatGPT to perform a systematic review: a tutorial. *Minerva Cardiology and Angiology* 72, 6 (Dec. 2024). https://doi.org/10.23736/s2724-5683.24.06568-2

[98] WM To and Billy TW Yu. 2023. Rise in higher education researchers and academic publications. *Emerald Open Research* 1, 3 (2023).

[99] Przemysław Tomczyk, Philipp Brüggemann, Niklas Mergner, and Maria Petrescu. 2024. Are AI tools better than traditional tools in literature searching? Evidence from E-commerce research. *Journal of Librarianship and Information Science* (Nov. 2024). https://doi.org/10.1177/09610006241295802

[100] Przemysław Tomczyk, Philipp Brüggemann, Niklas Mergner, and Maria Petrescu. 2024. *Exploring AI's Role in Literature Searching: Traditional Methods Versus AI-Based Tools in Analyzing Topical E-Commerce Themes*. Springer Nature Switzerland, 141–148. https://doi.org/10.1007/978-3-031-62135-2_15

[101] Angel Sebastian Treviño-Juarez. 2024. Assessing Risk of Bias Using ChatGPT-4 and Cochrane ROB2 Tool. *Medical Science Educator* 34, 3 (April 2024), 691–694. https://doi.org/10.1007/s40670-024-02034-8

[102] Guy Tsafnat, Paul Glasziou, George Karystianis, and Enrico Coiera. 2018. Automated screening of research studies for systematic reviews using study characteristics. *Systematic Reviews* 7, 1 (April 2018). https://doi.org/10.1186/s13643-018-0724-7

[103] Leonieke M. van den Bulk, Yamine Bouzembrak, Anand Gavai, Ningjing Liu, Lukas J. van den Heuvel, and Hans J.P. Marvin. 2022. Automatic classification of literature in systematic reviews on food safety using machine learning. *Current Research in Food Science* 5 (2022), 84–95. https://doi.org/10.1016/j.crfs.2021.12.010

[104] Max van Haastrecht, Injy Sarhan, Bilge Yigit Ozkan, Matthieu Brinkhuis, and Marco Spruit. 2021. SYMBALS: A Systematic Review Methodology Blending Active Learning and Snowballing. *Frontiers in Research Metrics and Analytics* 6 (May 2021). https://doi.org/10.3389/frma.2021.685591

[105] Siw Waffenschmidt, Elke Hausner, Wiebke Sieben, Thomas Jaschinski, Marco Knelangen, and Inga Overesch. 2018. Effective study selection using text mining or a single-screening approach: a study protocol. *Systematic Reviews* 7, 1 (Oct. 2018). https://doi.org/10.1186/s13643-018-0839-x

[106] Vickie R. Walker, Charles P. Schmitt, Mary S. Wolfe, Artur J. Nowak, Kuba Kulesza, Ashley R. Williams, Rob Shin, Jonathan Cohen, Dave Burch, Matthew D. Stout, Kelly A. Shipkowski, and Andrew A. Rooney. 2022. Evaluation of a semi-automated data extraction tool for public health literature-based reviews: Dextr. *Environment International* 159 (Jan. 2022), 107025. https://doi.org/10.1016/j.envint.2021.107025

[107] Byron C Wallace, Thomas A Trikalinos, Joseph Lau, Carla Brodley, and Christopher H Schmid. 2010. Semi-automated screening of biomedical citations for systematic reviews. *BMC Bioinformatics* 11, 1 (Jan. 2010). https://doi.org/10.1186/1471-2105-11-55

[108] Shuai Wang, Harrisen Scells, Bevan Koopman, Martin Potthast, and Guido Zuccon. 2023. Generating Natural Language Queries for More Effective Systematic Review Screening Prioritisation. In *Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP '23)*. ACM, 73–83. https://doi.org/10.1145/3624918.3625322

[109] Shuai Wang, Harrisen Scells, Bevan Koopman, and Guido Zuccon. 2023. Can ChatGPT Write a Good Boolean Query for Systematic Review Literature Search?. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '23)*. ACM, 1426–1436. https://doi.org/10.1145/3539618.3591703

[110] Florian Wätzold, Bartosz Popiela, and Jonas Mayer. 2024. Methodology for AI-Based Search Strategy of Scientific Papers: Exemplary Search for Hybrid and Battery Electric Vehicles in the Semantic Scholar Database. *Publications* 12, 4 (Dec. 2024), 49. https://doi.org/10.3390/publications12040049

[111] Emma Wilson, Florenz Cruz, Duncan Maclean, Joly Ghanawi, Sarah K. McCann, Paul M. Brennan, Jing Liao, Emily S. Sena, and Malcolm Macleod. 2023. Screening for in vitro systematic reviews: a comparison of screening methods and training of a machine learning classifier. *Clinical Science* 137, 2 (Jan. 2023), 181–193. https://doi.org/10.1042/cs20220594

[112] Tim Wittenborg, Oliver Karras, and Sören Auer. 2024. SWARM-SLR – Streamlined Workflow Automation for Machine-actionable Systematic Literature Reviews. (2024). https://doi.org/10.48550/ARXIV.2407.18657

[113] Phongphat Wiwatthanasetthakarn, Wanchana Ponthongmak, Panu Looareesuwan, Amarit Tansawet, Pawin Numthavaj, Gareth J McKay, John Attia, and Ammarin Thakkinstian. 2024. Development and Validation of a Literature Screening Tool: Few-Shot Learning Approach in Systematic Reviews. *Journal of Medical Internet Research* 26 (Dec. 2024), e56863. https://doi.org/10.2196/56863

[114] Anjia Ye, Ananda Maiti, Matthew Schmidt, and Scott J. Pedersen. 2024. A Hybrid Semi-Automated Workflow for Systematic and Literature Review Processes with Large Language Model Analysis. *Future Internet* 16, 5 (May 2024), 167. https://doi.org/10.3390/fi16050167

[115] Tom Young, Devamanyu Hazarika, Soujanya Poria, and Erik Cambria. 2018. Recent trends in deep learning based natural language processing. *IEEE Computational Intelligence Magazine* 13, 3 (2018), 55–75. https://doi.org/10.1109/MCI.2018.2840738

[116] Sanaz Zamani and Roopak Sinha. 2024. Generative AI — The End of Systematic Reviews in PhD Projects? *ACM Inroads* 15, 2 (May 2024), 48–50. https://doi.org/10.1145/3657304

[117] Eftim Zdravevski, Petre Lameski, Vladimir Trajkovik, Ivan Chorbev, Rossitza Goleva, Nuno Pombo, and Nuno M. Garcia. 2019. *Automation in Systematic, Scoping and Rapid Reviews by an NLP Toolkit: A Case Study in Enhanced Living Environments.* Springer International Publishing, 1–18. https://doi.org/10.1007/978-3-030-10752-9_1

[118] Marija Đukić, Milica Škembarević, Olga Jejić, and Ivan Luković. 2024. *Towards the Utilization of AI-Powered Assistance for Systematic Literature Review.* Springer Nature Switzerland, 195–205. https://doi.org/10.1007/978-3-031-70421-5_16

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B APPENDIX: FULL SEARCH QUERY & FULL DATA EXTRACTION TABLE

(TITLE-ABS-KEY ("systematic review") OR TITLE-ABS-KEY ("meta-analysis") OR TITLE-ABS-KEY ("meta analysis") OR TITLE-ABS-KEY ("systematic literature review") OR TITLE-ABS-KEY ("evidence synthesis")) AND (TITLE-ABS-KEY ("artificial intelligence") OR TITLE-ABS-KEY ("AI") OR TITLE-ABS-KEY ("machine learning") OR TITLE-ABS-KEY ("large language model*") OR TITLE-ABS-KEY ("LLM") OR TITLE-ABS-KEY ("natural language processing") OR TITLE-ABS-KEY ("ChatGPT") OR TITLE-ABS-KEY ("automation")) AND (TITLE-ABS-KEY (tool*) OR TITLE-ABS-KEY ("software") OR TITLE-ABS-KEY ("framework*") OR TITLE-ABS-KEY ("screening") OR TITLE-ABS-KEY ("data extraction") OR TITLE-ABS-KEY ("study selection")) AND NOT (TITLE-ABS-KEY ("randomized controlled trial") OR TITLE-ABS-KEY ("RCT") OR TITLE-ABS-KEY ("patient*") OR TITLE-ABS-KEY ("clinical") OR TITLE-ABS-KEY ("diagnos*") OR TITLE-ABS-KEY ("therap*") OR TITLE-ABS-KEY ("surg*") OR TITLE-ABS-KEY ("pharmaco*")
OR TITLE-ABS-KEY ("hardware") OR TITLE-ABS-KEY ("circuit*") OR TITLE-ABS-KEY ("sensor*") OR TITLE-ABS-KEY ("robot*") OR TITLE-ABS-KEY ("IoT")
OR TITLE-ABS-KEY ("curriculum") OR TITLE-ABS-KEY ("pedagog*") OR TITLE-ABS-KEY ("student*")
OR TITLE-ABS-KEY ("statistical method*") OR TITLE-ABS-KEY ("laboratory") OR TITLE-ABS-KEY ("experiment*")) AND NOT (TITLE-ABS-KEY ("covid*") OR TITLE-ABS-KEY ("corona") OR TITLE-ABS-KEY ("disease*") OR TITLE-ABS-KEY ("tumor*") OR TITLE-ABS-KEY ("cancer*") OR TITLE-ABS-KEY ("infection*"))

C RSTUDIO CODE

Listing 1. R code used for chi-square test of AI type vs PRISMA phases

```
data <- matrix(c(
  45, 53, 26, 6, 2, 5, # ML
  5, 10, 4, 1, 1, 0, # NLP
  7, 14, 5, 0, 1, 1, # LLM
  1, 0, 1, 0, 1, 0 # DL
),
nrow = 4, byrow = TRUE)

rownames(data) <- c("ML", "NLP", "LLM", "DL")
colnames(data) <- c("Identification", "Screening", "
     Data␣Extraction",
                 "Data␣Synthesis", "Quality␣Assessment
                      ", "Bias␣Assessment")


result <- chisq.test(data)
print(result)
```

Table 3. Complete Data Extraction of All Systematic Review Tools (128 Tools)

ID	Study	Year	Primary Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Im/Export	Biblio	Domain	Output	Effectiveness
31	David24	2024	[19]	NLP-based screening	framework	Screening, Data Extraction	NLP	Sentiment accuracy	Semi-automated	Standalone	Study data set	Open	No	No	Software engineering		Accuracy rate (86.5%)
83	Atkinson24	2024	[7]	SPARK	tool	Identification, Screening, Data Extraction	NLP (LDA)	Coherence score	Semi-automated	Standalone	Text	Open	Yes	No	Not specified	CSV	
138	Marshall24	2024	[95]	GPT-3 Prompt Templates	framework	Screening	LLM	Sensitivity score	Semi-automated	Standalone	Text	Open	No	No	Systematic reviews of RCTs	Text	Screening sensitivity (97.7%)
162	Ashleigh24	2024	[84]	DoCTER	tool	Screening	ML	Ensemble score	Semi-automated	Standalone	Text		No	No	Health	Text	
194	Chen24	2024	[17]	LLM-FA	tool	Screening	LLM	Accuracy score	Automated	Standalone	Text (prompts)	Open	No	No	Cross-disciplinary	Text	32.7%, 19.4%, 6.1%
194	Chen24	2024	[17]	LLM-SA	tool	Screening	LLM	Accuracy score	Semi-automated	Standalone	Text (papers)	Open	No	No	Cross-disciplinary	Text	82.7% (included), 92.2% (excluded)
206	Jensen24	2024	[66]	ChatGPT4-based screening	tool	Data Extraction	LLM	Accuracy score	Automated	Standalone	Text	Open	No	No	Not specified	Text	92.4% accuracy (5.2% false data)
320	Waltzod24	2024	[110]	Search Strategy	framework	Identification											
383	Dukic24	2024	[118]	Elicit	tool	Identification, Screening	LLM	Recall, precision	Automated	Integrated	Text	Closed	Yes	Yes	Software engineering	Text	88.3% precision
383	Dukic24	2024	[118]	SciSpace	tool	Identification, Screening	LLM	Recall, precision	Automated	Integrated	Text, PDF	Closed	Yes	Yes	Software engineering	Text	78.8% precision
392	Jiang24	2024	[48]	ChatGPT-based screening	tool	Identification, Screening, Data Extraction	LLM	Accuracy score	Semi-automated	Standalone	Text	Open	No	No	Agriculture	Text document	0.941
418	Lixia24	2024	[31]	Abstrackr	tool	Screening	ML										
418	Lixia24	2024	[31]	DistillerSR	tool	Identification, Screening, Data Extraction	ML				Title, abstract	Closed					
418	Lixia24	2024	[31]	RobotReviewer	tool	Data Extraction, Quality Assessment	NLP				Title, abstract	Closed					
418	Lixia24	2024	[31]	Rayyan	tool	Screening	ML				Abstract, title, keywords	Closed					
418	Lixia24	2024	[31]	EPPI-Reviewer	tool	Screening, Data Synthesis	ML				Abstract, title, keywords	Closed					
418	Lixia24	2024	[31]	ASReview	tool	Screening	ML				Text	Open					
418	Lixia24	2024	[31]	Covidence	tool	Screening	ML					Closed					
418	Lixia24	2024	[31]	SWIFT-Review	tool	Identification, Screening, Data Extraction, Data Synthesis	ML					Open					
418	Lixia24	2024	[31]	SearchRefiner	tool	Identification	ML				References	Open					
418	Lixia24	2024	[31]	Elicit	tool	Identification, Data Extraction, Data Synthesis	NLP				Title, abstract, PDFs	Closed					
418	Lixia24	2024	[31]	Nested Knowledge	tool	Identification, Screening, Data Synthesis	ML				Keywords	Closed					
418	Lixia24	2024	[31]	SWIFT-Active Screener	tool	Screening	ML				Abstract, title, keywords	Closed					
418	Lixia24	2024	[31]	Colandr	tool	Screening, Data Extraction	ML				Abstract, title, keywords	Closed					
418	Lixia24	2024	[31]	SciSpace GPT	tool	Data Extraction, Quality Assessment	DL				Papers	Closed					
418	Lixia24	2024	[31]	SciSpace	tool	Screening, Data Extraction	ML				Text	Closed					
451	Pilz24	2024	[78]	TIAB-Screener	tool	Screening	ML	Area under curve	Semi-automated	Standalone	Text	Open	No	No	Not specified		AUC table available

Continued on next page

Table 3 – continued

ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Im/Export	Biblio	Domain	Output	Effectiveness
543	Issaiy24	2024	[44]	ChatGPT-based screening	framework	Screening	LLM	Sensitivity score, AUC, PPV & NPV, CIs	Automated	Standalone	Text	Closed	No	No	Radiology		Sensitivity 95%, NPV 99%
550	Teperikidis24	2024	[97]	ChatGPT4-based screening	framework	Identification, Screening	LLM		Semi-automated	Standalone	Text	Closed	No	No	Not specified		
584	Menold24	2024	[62]	ML-enabled screening	framework	Screening	ML	Specificity, ROC curves, missed studies, work saved	Semi-automated	Standalone	Studies	Closed	No	No	Not specified		
589	Li24	2024	[56]	ChatGPT v4-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		96.1% accuracy, 96% specificity
589	Li24	2024	[56]	ChatGPT v3.5-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		94.5% accuracy, 87% specificity
589	Li24	2024	[56]	Google PaLM 2-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		90% accuracy, 95% specificity
589	Li24	2024	[56]	Llama2-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		78% accuracy, 61% specificity
589	Li24	2024	[56]	Gemini 1.0-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		87% accuracy, 99% specificity
589	Li24	2024	[56]	Llama3-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		91% accuracy, 89% specificity
589	Li24	2024	[56]	Claude3-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		92% accuracy, 94% specificity
629	Boateng24	2024	[72]	FastRead	tool												
629	Boateng24	2024	[72]	Research Screener	tool		ML		Semi-automated	Standalone	Titles	Open					
629	Boateng24	2024	[72]	Robot Analyst	tool												
634	Thomas24	2024	[68]	ChatGPT v3.5-based screening	tool	Screening	LLM	Accuracy score	Semi-automated	Standalone	Abstract, title	Closed	No	No	Ecological		83% accuracy
634	Liu24	2024	[58]	SWIFT-Active Screener	tool	Screening	ML	Agreement	Automated	Standalone	Abstract, title	Closed	No	No			97.9% agreement
721	Han24	2024	[36]	RAG-Based screening	framework	Identification, Screening, Data Extraction, Data Synthesis	LLM		Semi-automated	Standalone	Text	Closed	No	No	Not specified		
750	Adam24	2024	[1]	Literature search sandbox	tool	Identification	LLM	Sensitivity score	Semi-automated	Standalone	Text	Open	No	No	Not specified		85% median sensitivity
766	Fang24	2024	[26]	SyROCCo	tool	Screening, Data Extraction, Data Synthesis	ML	Similarity score	Automated	Standalone	Text	Closed	No	No	Not specified		0.711
863	Bolanos24	2024	[10]	Abstrackr	tool	Screening	ML										
863	Bolanos24	2024	[10]	Colandr	tool	Screening	ML										
863	Bolanos24	2024	[10]	Covidence	tool	Screening	ML										
863	Bolanos24	2024	[10]	EPPI-Reviewer	tool	Screening	ML										
863	Bolanos24	2024	[10]	Rayyan	tool	Screening	ML										
863	Bolanos24	2024	[10]	SWIFT-Active Screener	tool	Screening	ML										
863	Bolanos24	2024	[10]	DistillerSR	tool	Screening	ML										
863	Bolanos24	2024	[10]	SWIFT-REVIEW	tool	Screening	ML										
863	Bolanos24	2024	[10]	SysRev	tool	Screening	ML										
863	Bolanos24	2024	[10]	Nested Knowledge	tool	Screening	ML										
863	Bolanos24	2024	[10]	RobotAnalyst	tool	Screening	ML										
863	Bolanos24	2024	[10]	LitSuggest	tool	Screening	ML										
863	Bolanos24	2024	[10]	Pitts.ai	tool	Screening	ML										
863	Bolanos24	2024	[10]	Research Screener	tool	Screening	ML										
863	Bolanos24	2024	[10]	ASReview	tool	Screening	ML										
863	Bolanos24	2024	[10]	PICOPortal	tool	Screening	ML										
863	Bolanos24	2024	[10]	FAST2	tool	Screening	ML										
863	Bolanos24	2024	[10]	Dextr	tool	Data Extraction	ML										
863	Bolanos24	2024	[10]	ExaCT	tool	Data Extraction	ML										
863	Bolanos24	2024	[10]	Iris.ai	tool	Screening, Data Extraction	ML										

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ID	Study	Year	Ref	Tool Name	Type	PRISMA	Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Im/Export	Biblio	Domain	Output	Effectiveness
863	Bolanos24	2024	[10]	RobotReviewer/RobotSearch	tool	Screening, Extraction	Data	ML										
910	Landschaft24	2024	[54]	ChatGPT4-Based Screening	tool	Screening		LLM	Agreement, accuracy, sensitivity, specificity, F1-score	Automated	Standalone	Text	Closed	No	No	Healthcare		0.899, 0.976, 0.987, 0.950
1008	Butzke24	2024	[14]	SMAFIRA	tool	Identification		DL		Semi-automated	Standalone	Reference	Closed	No	No	Animal research		
1009	Spillias23	2023	[94]	ChatGPT-based screening	framework	Screening		LLM		Semi-automated	Standalone	Text	Closed	No	No			
1072	Syriani24	2024	[45]	ChatGPT-based screening	tool	Screening		LLM	Precision, NPV, specificity	Semi-automated	Standalone	Text	Closed	No	No			
1078	Ofori-Boateng24	2024	[71]	Covidence	tool	Screening, Extraction	Data			Semi-automated								
1078	Ofori-Boateng24	2024	[71]	JBI-SUMARI	tool	Screening, Extraction	Data			Semi-automated								
1078	Ofori-Boateng24	2024	[71]	Rayyan	tool	Screening												
1078	Ofori-Boateng24	2024	[71]	GRADEpro	tool													
1078	Ofori-Boateng24	2024	[71]	Nvivo	tool													
1078	Ofori-Boateng24	2024	[71]	Systematic Review Accelerator	tool	Screening, Extraction	Data			Semi-automated								
1078	Ofori-Boateng24	2024	[71]	Polygot	tool													
1078	Ofori-Boateng24	2024	[71]	Abstrackr	tool													
1078	Ofori-Boateng24	2024	[71]	DistillerSR	tool													
1078	Ofori-Boateng24	2024	[71]	Yale MESH Analyser	tool													
1078	Ofori-Boateng24	2024	[71]	MeSH on Demand	tool													
1078	Ofori-Boateng24	2024	[71]	EPPI-Reviewer	tool	Screening				Semi-automated								
1078	Ofori-Boateng24	2024	[71]	RobotAnalyst	tool	Screening				Semi-automated								
1078	Ofori-Boateng24	2024	[71]	RobotReviewer	tool													
1078	Ofori-Boateng24	2024	[71]	Atlas Ti	tool													
1078	Ofori-Boateng24	2024	[71]	PubReMiner	tool													
1078	Ofori-Boateng24	2024	[71]	Research Screener	tool	Screening				Semi-automated								
1078	Ofori-Boateng24	2024	[71]	RevTools	tool													
1078	Ofori-Boateng24	2024	[71]	SWIFT-ACTIVE Review	tool	Screening				Semi-automated								
1120	Chernikova24	2024	[18]	ML-enabled screening	tool	Screening		ML	Accuracy, recall, precision, F-score	Semi-automated	Standalone	Abstract, title	Closed	No	No			See table
1145	Hawkins24	2024	[49]	General-purpose filtering system	tool	Screening		NLP	Area under curve (AUC)	Semi-automated	Standalone	Abstract, title		No	No			See table
1163	Huotala24	2024	[43]	ChatGPT3.5-based screening	tool	Screening		LLM	Accuracy	Semi-automated	Standalone	Abstract, title		No	No			69
1239	Park24	2024	[116]	NLP-based screening	tool	Identification		NLP		Semi-automated	Standalone	Abstract, title		No	No			
1241	Trevino-Juarez24	2024	[101]	ChatGPT4-Based Screening	tool	Bias Assessment		LLM		Semi-automated	Standalone			No	No			
1334	Napoleao24	2024	[67]	ML-enabled screening	tool	Screening		ML	Recall, precision	Automated	Standalone	Abstract, title		No	No	System Engineering		74%, 15%

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ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Im/Export	Biblio	Domain	Output	Effectiveness
1337	Ye24	2024	[114]	Hybrid Workflow	framework	Identification, Screening, Data Extraction	ML	Accuracy	Semi-automated	Standalone	Abstract, title, full text						100% accuracy
1415	Demir24	2024	[21]	ChatGPT-based screening	framework	Identification	ML	Accuracy	Semi-automated	Standalone	Abstract, title		No	No			See table
1478	Quan24	2024	[81]	ASReview	tool	Screening	ML		Semi-automated		Dataset		Yes	No			
1499	Campos24	2024	[15]	Screening Smarter not Harder	tool	Screening	ML	Recall, sensitivity, specificity, time saving	Semi-automated	Standalone	Papers		No	No			
1508	Pijls24	2024	[77]	ML-enabled screening	tool	Identification	ML	Recall, confidence interval	Semi-automated	Standalone	Text		No	No	Orthopedics		
1725	Wiwatthanasettharn24	2024	[113]	FSL-based screening	framework	Screening	ML	Workload reduction, cosine similarity, recall needs	Automated		Abstract, title	Closed	No	No			51.11% workload reduction
1734	Bruggemann24	2024	[99]	Elicit	tool	Identification, Screening	ML										
1734	Bruggemann24	2024	[99]	SciSpace	tool	Identification, Screening	ML										
1860	Quan24	2024	[80]	Abstrackr	tool	Screening	ML				Abstract, title				Second language acquisition		
1860	Quan24	2024	[80]	ASReview	tool	Screening	ML				Abstract, title						
1860	Quan24	2024	[80]	Rayyan	tool	Screening	ML				Abstract, title						
1980	Wittenborg24	2024	[112]	SWARM-SLR	tool	Identification, Screening, Data Extraction, Data Synthesis, Bias	AI		Semi-automated	Integrated		Open	Yes	No	General	SLR	
2115	Tomczyk24	2024	[100]	Elicit	tool	Identification, Screening	ML										
2115	Tomczyk24	2024	[100]	SciSpace	tool	Identification, Screening	ML										
2248	Luo24	2024	[59]	EPPI-Reviewer	tool	Screening	ML										
2248	Luo24	2024	[59]	DistillerSR	tool	Screening	ML										
2248	Luo24	2024	[59]	ChatGPT-based screening	tool	Identification, Screening, Data Extraction	LLM										
2248	Luo24	2024	[59]	Claude3-based screening	tool	Identification, Screening, Data Extraction	LLM										
2258	Oami24	2024	[70]	ChatGPT4 Turbo-Based Screening	tool	Screening	LLM	Sensitivity, specificity	Automated	Standalone	Abstract, title	Closed	No	No	Japanese sepsis guidelines		Sensitivity 0.75, specificity 0.9
2304	Soufi24	2024	[93]	Chatpdf	tool	Data Extraction	ML		Automated	Integrated	PDF	Closed	Yes	No			
2304	Soufi24	2024	[93]	Pdf2gpt	tool	Data Extraction	ML		Automated	Integrated	PDF	Closed	Yes	No			
2304	Soufi24	2024	[93]	Hipdf	tool	Data Extraction	ML		Automated	Integrated	PDF	Closed	Yes	No			
2304	Soufi24	2024	[93]	SciSpace	tool	Screening, Data Extraction	ML		Semi-automated	Standalone	Text	Closed	No	No			
2304	Soufi24	2024	[93]	Easy-peasy AI	tool												
2304	Soufi24	2024	[93]	DocAnalyzer AI	tool	Screening, Data Extraction	ML		Semi-automated	Standalone	Text	Closed	No	No			
2343	Ruksakulpiwat24	2024	[88]	ChatGPT-based screening	tool	Identification, Screening	LLM	Precision	Semi-automated	Standalone	Text	Closed	No	No	Health		0.86
2422	Hasan24	2024	[38]	ChatGPT4-Based Screening	framework	Bias	LLM	Risk of bias	Semi-automated	Integrated	Text	Closed	No	No			
2463	Khan24	2024	[83]	ChatGPT4-Turbo-Based Screening	tool	Data Extraction	LLM	Accuracy	Automated	Standalone	Text	Closed	No	No	Clinical trials		
2463	Khan24	2024	[83]	Claude3-based screening	tool	Data Extraction	LLM	Accuracy	Automated	Standalone	Text	Closed	No	No	Clinical trials		
2583	Guo23	2023	[34]	ChatGPT-Based Screening	tool	Screening	LLM	Accuracy, sensitivity, F1-score	Automated	Standalone	Abstract, title	Closed	No	No	See table		See table
2583	Guo23	2023	[34]	ChatGPT4-Based Screening	tool	Screening	LLM	Accuracy, sensitivity, F1-score	Automated	Standalone	Abstract, title	Closed	No	No	See table		

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ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Im/Export	Biblio	Domain	Output	Effectiveness
2709	Quershi23	2023	[82]	ChatGPT3.5-based screening	tool	Screening	LLM		Automated	Standalone	Abstract, title	Closed	No	No			
2715	Qang23	2023	[108]	Natural Language Query's	tool	Identification	NLP	Recall, accuracy	Semi-automated	Standalone	Text	Closed	No	No			See table
2737	Kusa23	2023	[53]	CRUISE-Screening	tool	Screening	ML		Automated	Integrated		Open	Yes	Yes			
2883	Roberts23	2023	[65]	ChatGPT3-based screening	tool	Identification	LLM	Mean difference	Automated	Standalone	Text	Closed	No	No			4.92% difference
2933	Roth23	2023	[86]	RCT tagger	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	Litsearchr	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	WordNet	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	RobotSearch	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	Zettair	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	SLR.qub	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	TerMine	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	Leximancer	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	Paperfetcher	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	Deducklick	tool	Identification	ML		Automated								
2933	Roth23	2023	[86]	DistillerSR	tool	Screening	ML		Automated								
2933	Roth23	2023	[86]	Rayyan	tool	Screening	ML		Automated								
2933	Roth23	2023	[86]	Colandr	tool	Screening	ML		Automated								
2933	Roth23	2023	[86]	Abstrackr	tool	Screening	ML		Automated								
2933	Roth23	2023	[86]	EPPI-Reviewer	tool	Screening	ML		Semi-automated								
2933	Roth23	2023	[86]	ASReview	tool	Screening	ML		Automated								
2933	Roth23	2023	[86]	RobotAnalyst	tool	Screening	ML		Semi-automated								
2933	Roth23	2023	[86]	SWIFT-Review	tool	Screening, Data Extraction	ML		Semi-automated								
2933	Roth23	2023	[86]	MetaMap	tool	Screening	ML		Automated								
2933	Roth23	2023	[86]	RapidMiner	tool	Screening	ML		Automated								
2933	Roth23	2023	[86]	SyRF	tool	Screening	ML		Automated								
2933	Roth23	2023	[86]	PDFBox	tool	Data Extraction	ML		Automated								
2933	Roth23	2023	[86]	GATE	tool	Data Extraction	ML		Automated								
2933	Roth23	2023	[86]	EXACT	tool	Data Extraction	ML		Automated								
2933	Roth23	2023	[86]	RobotReviewer	tool	Data Synthesis	ML		Automated								
2933	Roth23	2023	[86]	Robumeta	tool	Bias	ML		Automated	Integrated							
2933	Roth23	2023	[86]	PublicationBias	tool	Bias	ML		Automated	Integrated							
2933	Roth23	2023	[86]	2weekSR	tool	Identification, Screening, Data Extraction	ML		Semi-automated								
3037	Santos23	2023	[89]	Abstrackr	tool	Screening	ML		Semi-automated								
3037	Santos23	2023	[89]	Active Pubmed Search (APS)	tool	Identification	ML		Semi-automated								
3037	Santos23	2023	[89]	Cochrane RCT classifier	tool	Screening	ML		Automated						RCTs		
3037	Santos23	2023	[89]	Colandr	tool	Identification	ML: NLP		Semi-automated								
3037	Santos23	2023	[89]	DistillerSR	tool	Screening	ML		Automated								
3037	Santos23	2023	[89]	EPPI-Reviewer	tool	Identification, Screening	ML										
3037	Santos23	2023	[89]	ExaCT	tool	Data Extraction	ML: Text Mining		Automated								
3037	Santos23	2023	[89]	GAPScreener	tool	Screening	ML		Semi-automated						Human genetics		
3037	Santos23	2023	[89]	IBM Parse	tool	Identification	ML: Text Mining		Semi-automated								
3037	Santos23	2023	[89]	LitSuggest	tool	Identification	ML		Semi-automated								
3037	Santos23	2023	[89]	MetaPreg	tool	Identification, Screening, Data Extraction	ML: NLP		Semi-automated						Pregnancy meds		

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ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Im/Export	Biblio	Domain	Output	Effectiveness
3037	Santos23	2023	[89]	MMiDas-AE	tool	Screening	NLP		Semi-automated								
3037	Santos23	2023	[89]	NaCTeM	tool	Data Extraction	ML: Text Mining		Automated						Genes		
3037	Santos23	2023	[89]	Rayyan	tool	Screening	ML		Semi-automated								
3037	Santos23	2023	[89]	ResearchScreener	tool	Screening	ML		Semi-automated								
3037	Santos23	2023	[89]	Revtools	tool	Data Synthesis	NLP: Text Mining		Automated								
3037	Santos23	2023	[89]	RobotAnalyst	tool	Screening	ML		Automated								
3037	Santos23	2023	[89]	RobotReviewer	tool	Data Extraction, Bias	ML		Semi-automated						RCTs		
3037	Santos23	2023	[89]	RobotSearch	tool	Identification	ML: NLP										
3037	Santos23	2023	[89]	Screen4Me	tool	Identification	ML								RCTs		
3037	Santos23	2023	[89]	SRToolBox	tool	Identification, Screening, Data Extraction, Data Synthesis	ML		Automated								
3037	Santos23	2023	[89]	SWFIT-Active Screeer	tool	Screening	ML		Automated								
3037	Santos23	2023	[89]	SWFIT-Review	tool	Identification	ML		Automated								
3037	Santos23	2023	[89]	Twister	tool	Identification	ML: Text Mining		Semi-automated								
3038	Alshami23	2023	[3]	ChatGPT3.5-based screening	tool	Identification, Screening, Data Extraction, Data Synthesis	LLM	Classification, F1-score	Automated	Standalone	Text	Closed	No	No		Text	91% classification, 88% F1
3060	Wang23	2023	[109]	ChatGPT-based screening	tool	Identification	LLM	Precision, recall, F1-score, F3-score	Semi-automated	Standalone	Text	Closed	No	No		Text	
3157	Scott23	2023	[91]	2weekSR	tool	Identification, Screening, Data Extraction	ML		Semi-automated	Integrated							
3342	Manning23	2023	[46]	Elicit	tool	Identification, Screening	ML		Semi-automated	Standalone	Text (prompt)	Closed				Text	
3342	Manning23	2023	[46]	ResearchRabbit	tool	Identification, Screening	NLP		Semi-automated	Standalone	Text (prompt)	Closed				Text	
3621	Sundaram23	2023	[96]	NLP/TM-based screening	tool	Data Extraction, Data Synthesis	NLP: Text Mining		Semi-automated	Standalone							
3679	Faria23	2023	[27]	Transformer	tool	Screening	ML	Workload saved (WSS@R)	Semi-automated	Standalone	Papers						0.8
3780	Hasny23	2023	[39]	srBERT	tool	Screening	ML: NLP	Recall, F1-score, screening reduction	Automated	Integrated	Abstract, titles	Open	No	No		Text	See table
3814	Wilson23	2023	[111]	In-Vitro screening	tool	Screening	ML: Text Mining	Sensitivity, specificity, precision	Automated	Standalone	Abstract, titles, full text	Closed	No	No		Text	0.972, 0.980, 0.764
3993	O'Keefe23	2023	[74]	Anne O'Tate	tool	Identification	ML: Text Mining		Semi-automated	Integrated	Search string (text)					Tabular	
3993	O'Keefe23	2023	[74]	BiblioShiny	tool	Identification	ML: Text Mining		Semi-automated	Integrated	Scopus/WoS/PubMed file		Yes			Visualizations	
3993	O'Keefe23	2023	[74]	Carrot2	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Search string (text)		Yes			Visualizations	
3993	O'Keefe23	2023	[74]	CitNetExplorer	tool	Identification	ML: Text Mining		Semi-automated	Standalone	WoS file		Yes			Network maps	
3993	O'Keefe23	2023	[74]	EndNote	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Papers		Yes			Tabular	
3993	O'Keefe23	2023	[74]	Keyword-Analyzer	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Text		Yes			Tabular	
3993	O'Keefe23	2023	[74]	Lingo3G (Carrot)	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Search string (text)		Yes			Visualizations	
3993	O'Keefe23	2023	[74]	Lingo4G	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Search string (text)		Yes			Visualizations	

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ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Im/Export	Biblio	Domain	Output	Effectiveness
3993	O'Keefe23	2023	[74]	MeSHonDemand	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Full text		Yes			Tabular	
3993	O'Keefe23	2023	[74]	PubReMiner	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Search string (text)		Yes			Tabular	
3993	O'Keefe23	2023	[74]	TerMine	tool	Identification	ML: 3993	O'Keefe23	2023	[74]	Text Analyzer	tool	Identification	ML: Text Mining		Semi-automated	Standalone
Text		Yes			Tabular												
3993	O'Keefe23	2023	[74]	Tm for R	tool	Identification	ML: Text Mining		Semi-automated	Integrated	Papers (PDF)		Yes			Tabular	
3993	O'Keefe23	2023	[74]	VosViewer	tool	Identification	ML: Text Mining		Semi-automated	Standalone	DOI/JSON/Scopus		Yes			Network maps	
3993	O'Keefe23	2023	[74]	Voyant	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Full text		Yes			Multiple visualizations	
3993	O'Keefe23	2023	[74]	Yale MeSH Analyser	tool	Identification	ML: Text Mining		Semi-automated	Standalone	PubMed IDs		Yes			Tabular	
3994	Ammirato22	2022	[5]	MySLR	tool	Identification, Screening, Data Extraction, Data Synthesis	ML: Text Mining, NLP		Semi-automated	Integrated							
4117	Adam22	2022	[2]	Pythia	tool	Screening	ML: NLP	Precision, sensitivity	Semi-automated	Standalone	Abstract, titles		No	No			Sensitivity 5[illegible]%
4174	Facchinetti22	2022	[24]	slr-kit	framework	Data Extraction, Bias, Analysis, QA	ML	Precision, recall	Semi-automated	Integrated	Abstract, title		Yes	Yes		Papers	See table
4387	Bernardes22	2022	[9]	ASReview	tool	Screening	ML		Semi-automated	Standalone	Abstract, title				Toxicology		
4413	Walker22	2022	[106]	Dextr	tool	Data Extraction	ML	Precision, recall	Semi-automated	Standalone	Full text	Closed	Yes	No	Public health	Text	96%, 91.8%
4730	Bulk22	2022	[103]	ML-based screening	tool	Screening	ML	Precision, recall, F1-score	Automated	Standalone	Abstract, title	Closed	No	No	Food consumption	Classification	57.8% irrelevant reduction
4746	Hannousse22	2022	[37]	Document Screener	tool	Screening	ML	F1-score, accuracy	Semi-automated	Standalone	Metadata	Open	No	No		Classification	89% F1, 97% accuracy
4782	Kusa22	2022	[52]	Citation Screener	tool	Screening	ML	Included, excluded, WSS	Automated	Standalone	Citations		No	No	Multiple		Dataset dependent
4842	Aum21	2021	[8]	srBert	tool	Screening	ML	Accuracy, F1-score, AUC	Automated	Standalone	Abstract, title	Closed	No	No		Papers	94.35% accuracy
4893	Holub21	2021	[40]	AutoLit	tool	Data Extraction	ML		Semi-automated				No	No			
4901	Chai21	2021	[16]	Research Screener	tool	Screening	ML	Workload saving	Semi-automated	Standalone	Abstract, title		Yes	No			60-96% across SLRs
5156	Noel-Storr21	2021	[69]	Screen4Me	tool	Screening	ML	Agreement, sensitivity, specificity	Semi-automated	Integrated	Abstract, title		Yes	No		Classification	100% sensitivity
5275	Haastrecht21	2021	[104]	SYMBALS	tool	Screening	ML: Active Learning	Speed	Automated	Standalone	Abstract, title	Open	No	No			6x faster
5525	Dhrangadhariya20	2020	[6]	Domain specific Citation Screening	tool	Screening	ML	Precision, recall, F1, AUC	Automated	Standalone	Abstract, titles		No	No	Multiple		Algorithm dependent
5565	Pradhan19	2019	[79]	EXACT	tool	Data Extraction	AI: Data Mining		Semi-automated	Standalone	Full text		Yes	Yes		Spreadsheet	
5579	Gates20	2020	[29]	Abstrackr	tool	Screening	ML	Precision, time saved	Semi-automated	Standalone	Abstract, titles		Yes	No	Multiple	Tabular	3-82 hours saved
5822	Marshall19	2019	[60]	RobotSearch	tool	Identification	ML		Automated								
5822	Marshall19	2019	[60]	Cochrane Register	tool	Identification	ML										
5822	Marshall19	2019	[60]	RCT Tagger	tool	Identification	ML										
5822	Marshall19	2019	[60]	Thalia	tool	Identification	ML										
5822	Marshall19	2019	[60]	Abstrackr	tool	Screening	ML										
5822	Marshall19	2019	[60]	EPPI Reviewer	tool	Screening	ML										
5822	Marshall19	2019	[60]	RobotAnalyst	tool	Screening	ML										
5822	Marshall19	2019	[60]	SWIFT-Review	tool	Screening	ML										
5822	Marshall19	2019	[60]	Colandr	tool	Screening	ML										
5822	Marshall19	2019	[60]	Rayyan	tool	Screening	ML										
5822	Marshall19	2019	[60]	ExACT	tool	Data Extraction	ML										
5822	Marshall19	2019	[60]	RobotReviewer	tool	Data Extraction	ML										

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Table 3 – continued

ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Im/Export	Biblio	Domain	Output	Effectiveness
5822	Marshall19	2019	[60]	NaCTeM	tool	Data Extraction	ML: Text Mining										
5822	Marshall19	2019	[60]	RobotReviewer	tool	Bias	ML										
5870	Schmidt19	2019	[90]	RAPTOR: RevMan	tool	Data Extraction	ML		Automated	Integrated	Full text	Open	Yes	No		XML	
5897	Lerner19	2019	[55]	Automatic NMA Screener	tool	Screening	ML: NLP	Sensitivity, workload decrease	Automated	Standalone	Citations	Open code	No	No	Medical	Classification	100% sensitivity
5898	Martin19	2019	[61]	Trial2rev	tool	Identification, Screening, Data Extraction	ML		Semi-automated	Standalone		Open	Yes	Yes			
5906	Hu18	2018	[42]	MetaCyto	tool	Data Extraction	ML		Automated	Standalone	Full text		No	No	Cytometry		
5916	Waffenschmidt18	2018	[105]	Abstrackr	tool	Screening	ML		Semi-automated	Standalone							
5916	Waffenschmidt18	2018	[105]	Rayyan	tool	Screening	ML		Semi-automated	Standalone							
5916	Waffenschmidt18	2018	[105]	EPPI Reviewer	tool	Screening	ML		Semi-automated	Integrated							
5957	Zdraveski19	2019	[117]	NLP Toolkit	tool	Identification, Screening	ML: NLP		Automated	Integrated	Abstract, titles	Open	Yes	No	Assisted living	Multiple	
5964	Goswami19	2019	[32]	Data Extracter	tool	Data Extraction	ML	Recall, precision	Automated	Standalone	Abstract, titles		No	No	Anxiety measures	Sentences	100% recall
6006	Kreiner18	2018	[50]	Twister	tool	Screening	ML: Text Mining		Semi-automated	Standalone	Abstract, titles	Open	Yes	No	Clinical guidelines	PDF	
6090	Gates18	2018	[30]	Abstrackr	tool	Screening	ML	Sensitivity, specificity, precision	Semi-automated	Standalone	Abstract, titles	Closed	Yes	No	Health		>75% sensitivity
6098	Tsafnat18	2018	[102]	KeyCharacteristic Screener	tool	Screening	ML: Text Mining	Recall, exclusion	Automated	Standalone	Abstract, titles	Closed	No	No			93.7% excluded
6132	Howard16	2016	[41]	SWIFT-Review	tool	Screening	ML	WSS@R	Semi-automated	Standalone	Abstract, titles		Yes	Yes			Dataset dependent
6331	Bowes12	2012	[12]	SLuRp	tool												
6369	Garcia14	2014	[28]	Text-Classifier	tool	Screening	ML	Precision, recall	Semi-automated	Standalone	Abstract, titles				Medicine	Classification	Up to 84%
6421	Wallace10	2010	[107]	Citation Screener	tool	Screening	ML	Screening reduction	Semi-automated	Standalone	Abstract, titles				Biomedical	Citations	50% reduction